

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR,ANDERI WEST,MUMBAI.		
PART F: FINANCIAL BID		
SL. NO	DESCRIPTION	AMOUNT
1	CIVIL WORKS	92,28,46,030
2	ELECTRICAL WORKS + CCTV	9,96,67,551
3	PLUMBING WORKS + PUMP ROOM	5,99,66,700
4	FIRE FIGHTING + FAS & PA SYSTEM WORKS + FIRE EXTINGUISHER	2,24,05,245
5	LIFT WORK	1,51,42,000
6	GRAND TOTAL	1,12,00,27,526

Name of the Contractor / Company Name					
SN	Description	Estimated Cost (₹)	Percentage above or below the estimated cost	% in Figures	Total Cost
1	PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR,ANDERI WEST,MUMBAI.	1,12,00,27,526.00	Select		-
	Grand Total				

CONSTRUCTION OF 122 NOS RESIDENTIAL FLATS for SBI ,DN NAGAR, ANDHERI WEST MUMBAI					
PART F: BILL OF QUANTITIES - CIVIL WORKS					
Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
	Preamble				
	Rate shall be inclusive of following;				
i	The serviceable and unserviceable materials shall be stacked separately and properly in the designated place till the disposal / reuse of the same. The debris, excess materials is to be carted outside the premises and disposed off at the Municipal Corporation designated place up to any lead, lift and level by the Contractor at his own cost. The Rates inclusive of Transpotation and Labour				
ii	All the civil work has to be carried with minimum 53 Grade OPC cement and VSI grade crush sand, of required fineness, which must be free from stilt and other impurities				
iii	All the civil work , whether new or repair, must be cured for at least 7 days preferably by ponding using gunny bags.				
iv	For all work, water used must be fresh, potable and pure without any salinity and turbidity. Water test shall be carried out before commencement of works by the Contractor at his own cost and test results to be submitted to the SBI/APMC.				
v	The rates quoted for all the items of the BOQ shall include the following specifications: -				
a	All the works are to be carried out at all heights and locations i.e. from foundation level, Ground Floor of the building up to the Over Head Water Tank roof slab or Mumty roof slab, as applicable.				
b	Providing and fixing double scaffolding system (cup lock type) on the exterior side, up to any height (entire height) of the building structure made out of 40 mm dia. M.S. tubes, 1.5 m centre to centre, horizontal & vertical tubes joining with cup & lock system with M.S. tubes, M.S. clamps and M.S. staircase system in the scaffolding including the working platform etc. and maintaining it in a serviceable condition for the entire required duration as approved and directed by the SBI/APMC & removing it thereafter.				
c	The scaffolding system shall be stiffened with bracings, runners, connection with the building fascia etc. wherever required & including providing vertical supports which shall be sound and strong, tied together with horizontal steel / metal support pieces over which scaffolding planks shall be fixed properly & secured at required locations including the essential safety features for the workmen etc. complete as per directions and approval of SBI/APMC. Note: - The scaffolding shall not be removed without prior approval of SBI/APMC.				
d.	The quoted rate should include cost of necessary scaffolding, staging, centring, shuttering etc. as required for execution of the job at any height and location including removal of same. The scaffolding should be erected and anchored into the concrete members with help of expandable bolts, which are to be cut as the scaffolding is removed, no wall is to be punctured, at no extra cost to the Bank.				
e.	Providing & fixing temporary support arrangements such as Steel Props (Up to any number whenever and wherever as directed by the SBI/APMC) to support the structure provisionally during the project duration for all trades of work, etc. and maintaining the props in position till required & as directed by the consultant including other safety elements such as bracings, steel angles, working platforms, plates etc., which shall be provided & considered in the quoted rates for the areas in the vicinity in adequate numbers as directed. All trades of works indicated herein under shall include all leads & lifts & working up to any heights, depths & levels.				
f.	The Contractor will be responsible or arranging "design of concrete mix" through approved Government Engineering College / IIT/ NIT or any other institutes recommended by APMC/ SBI at their own cost within the quoted rates only and will submit the report to SBI/APMC for approval. No extra payment shall be eligible on this account.				

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
g.	Providing & fixing safety net around the work so that loose plaster/ concrete does not cause damage to the property or any human being, if necessary and removing the same after completion of work.				
h	Providing and covering entire external surface with Hessian cloth curtains through out the duration of project and carefully removing the same after completion of work.				
i	Providing all type of PPE (Personal Protective Equipments) it is mandatory to the workforce such as safety Nets all-round at all levels ,safety shoes, safety helmets, reflective safety jacket, full body safety harness, goggles, gloves, etc. complete for all types of workforce engaged on site for execution of the work.				
j	Providing and maintaining Barricading on the site along the boundary wall at 6 mts high by using MS Pipes or angle iron frame work of verticals at 3 m c/c and horizontals at 0.90 m c/c, GI Precoated corrugated 0.5mm thick sheets, Struts, MS Clamps, etc. The verticals shall be embedded/ fixed to concrete pedestals below ground. Including steel for foundation etc., painting the same with one coat primer & two coats of approved anti rust paint etc. complete. Contractor to get structural design approved by the Engineer. Contractor shall remove and take away the barricading at his own cost on completion of the works in parts as directed .				
k.	Rates shall be inclusive of all enabling works like Site office including all furniture and necessary requirements with MEP works for client and PMC ,tower cranes all safety measures, aerial lift, man & material lift, Temporary Passenger lift, scaffolding, Cup lock system, labour camp, storages, godowns, all required permissions from local governing respective authorities to carry out the works, all insurances, all licenses, administration, all S.H.E. related measures, security , watch and ward measures, barricades other than specified, traffic management, tree protection, property damage repairs, water, electricity, DG as per requirement, all tools, machineries, vehicles, etc. Complete.				
l	Before starting of Excavation NGL levels shall be got verified and approved by SBI/APMC and RL levels to be at the top of different layer and based on the difference levels excavation qty shall be calculated.				
m	Rates should be inclusive all loading, unloading, shifting, all leads, lifts, sorting, transportation, labours, tools, plants, machinery,etc. complete				
n	Disposing and carting away of the excavated surplus soil / Hard murum Hard earth rock from site as directed and dumping to a place approved by Mumbai Municipal Corporation. / yard including all the costs for getting permission from MMC etc. complete as directed. Including provision.				
o	All the above-mentioned points including the requisite elements & items as mentioned shall be considered included within the quoted rates of the line items of the BOQ and nothing extra would be paid on this account.				
I	EARTH WORK				
1	Clearing jungle including uprooting of rank vegetation, grass, brush wood, trees and saplings of girth and removal of rubbish for any depth etc. the rate inclusive of disposing the same to approved Municipal dumping area including cost of all lead, lift, loading, unloading and transportation complete as directed by the Bank. Rate inclusive of Tree cutting permission from local Authority.	Sqm	5,547	20	1,10,940

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
2	Excavation in all kinds of soil, including soft murum & hard murum				
	Excavation in all kinds of soil, including soft murum, hard murum or old foundation etc., wet or dry, in loose and slushy condition including clearing the site removing shrubs, tree roots, rank vegetation etc., to the required level for foundations, trenches, pits, rafts, plinths, plinth area levelling, etc.; to required depth from existing ground level including stacking suitable earth for backfilling within site and refilling the same in sides of foundation/ retaining wall, open areas and plinth and disposing off surplus excavated material within the site as per direction of the SBI/APMC at no extra cost; using mechanical/ electrical means such as poclain, JCB, rock breaker, chisel tools, compressor, dumper, dozers etc., bailing/pumping out water with required capacities dewatering pump to keep the site dry at all times, including making temporary stepping / sloping, shoring with planking, strutting, sectioning, trimming, dressing of sides, levelling or grading including sprinkling of water cleaning, removing and excavating loose pockets and consolidating them by backfilling as directed.				
2a	From 0.00 to 1.50 m depth	Cum	5,279	410	21,64,390
2b	From 1.50 to 3.00 m depth	Cum	490	749	3,67,010
2c	From 3.00 to 4.50 m depth	Cum	100	1,100	1,10,000
3	EXCAVATION IN ALL KINDS OF SOIL, INCLUDING SOFT ROCK, HARD ROCK OR OLD FOUNDATION				
	Earth work in excavation in all kinds of soft rock and Hard Rock by mechanical means (Hydraulic excavator) / manual means by chiselling, wedging, line drilling etc. over areas including foundations, trimming & levelling the bed, removing excavated material beyond building area upto 2.0 m depth, stacking the material as directed including dewatering, back filling of trenches using available earth murum ramming and watering complete and disposing off surplus rock material within the site as per direction of the Bank. Rate to include all lead, lift and cost of transportation within the campus/premises/site. Blasting is not permitted.				
a	Above 4.50 m depth	Cum	10	1,500	15,000
4	FILLING				
	Filling and back filling in foundations, trenches, plinths, area levelling				
	Filling and back filling in foundations, trenches, plinths, area levelling etc. in layers not exceeding 200mm in depth, breaking clods, watering, compacting each layer with vibrator/ compactor and at inaccessible places with wooden/steel rammers at optimum moisture content; dressing up embankments; if required, bailing/pumping out of water to keep site dry while back filling; cost includes conveyance of all materials, labour, machinery, mechanical mean such as dumpers, dozers, poclain/JCB for lifting, etc. taking permission of local authorities and paying royalty etc. complete as per direction of the SBI/APMC complete. RL shall be taken for before laying of Murum and after completion of Laying and compaction of Murum and the difference in Levels shall be multiplied by Plan area to arrive on the actual qty after making necessary deductions,				
4a	Filling with good quality earth /murum from available / excavated earth approved by the SBI/APMC. Rates are inclusive of transportation, watering, ramming to achieve compacted level all complete.	Cum	2,487	210	5,22,270
4b	Filling with good quality murum brought from outside approved by the SBI/APMC. Rates are inclusive of transportation watering, ramming to achieve compacted level complete.	Cum	5,953	1,600	95,24,800

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
5	QUARRY STONE DUST FILLING				
	Supplying and filling with Quarry Stone Dust in layers (each layer should not exceed 15 cm), having total minimum compacted thickness 30 cm or as decided by Engineer -in- charge, including compacting each layer by ramming and watering, all complete as per drawing and direction of Engineer -in - charge.	Cum	430	2,500	10,75,000
	TOTAL OF SECTION - I				1,38,89,410
II	PRE-CONSTRUCTIONAL ANTI-TERMITE TREATMENT				
6	Providing pre-constructional anti-termite treatment by treating the bottom surface & sides of excavation in soil with plinth beams, plinth filling and periphery confirming as per IS : 8944 & IS : 6313 Part-II 1981 or any latest revision and furnishing 10 years guarantee in required proforma all complete. (Reference Specification- IS-6313 Part-II of 1981 and IS-IS-8944- 1978) Chemical used- Chemical name Trade Name Manufacturer Indian Standard Concentrations Chlorpyriphos 20% EC Cologuard Gharda Chemicals IS-8944-year-1978 1% Imidacloprid 30.5% SC Premise Bayer CIB 0.075% Treatment for Buildings with basement- The treatment will start after site survey, when the excavation is completed and before doing soling and PCC. 1. Treatment to Soil below. The excavated area will be treated with chemical emulsion @ 5liter/ Sq mtr, before soling and PCC, the side wall of the pit up to height of 30 cm will also treated with chemical emulsion. 2. Treatment to Soil along the retaining wall- The soil in immediate contact with retaining wall will be treated with chemical emulsion @ 5 liter /Sq mtr. The treatment will be done to back filling to facilitate continuity of chemical barrier. Rodding will have carried out if required to facilitate chemical absorption.				
	3. Treatment of soil Along the External Perimeter of Building- The External perimeter is treated by making holes with Iron bar @ 5 liter/running metre. The iron rod moved backward & forward to break up the earth and chemical emulsion is poured into the holes to establish continuity of chemical barrier. 4. Treatment to the Expansion Joints- The expansion joints are also treated to avoid termite entry from these expansion joints which are treated with the chemical emulsion @ 5 liter/running metre. 5. Treatment of the soil surrounding pipes and conduits- This is also one of the important area from where the termite get easily entry, this area is treated @ 5liter/Sq mtr or linear mt. of chemical eulsion thoroughly, the total surrounding area is covered by pouring the chemical emulsion.(Approved vendors -PCI, Godrej, Bayer & Hicare)	Sqm	3,400	200	6,80,000
	TOTAL OF SECTION - II				6,80,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
	PREAMBLE : The Rates Quoted shall be inclusive of the following for all Concrete/RCC work:				
a	All shuttering material used for concreting / RCC Work should be of good quality using BWP grade plywood to ensure smooth finish				
b	The scope of work under the items includes execution of work at all heights and location including all shapes, curved walls etc. Complete.				
c	In case there is bulging / honey combing is noticed after removal of shuttering, the contractor will have to rectify the work at his own cost. The decision as to what constitute minor honey combing is left to the SBI/APMC.				
d.	The Contractor will be responsible or arranging "design of concrete mix" through Government Engineering College / IIT/ NIT or any other institutes recommended by APMC/ SBI at contractor's own cost within the quoted rates only and will submit the report to SBI/APMC for approval. No extra payment shall be eligible on this account.				
e.	The flooring, raft & slab curing is to be done by ponding method only min 7 days to be done.				
f.	Checking of top level of slab should be done at every floor & it should be within IS tolerance limit / client requirement.				
g.	Frequency of testing for all materials like sand, aggregates, cement, steel. Concrete should be as per IS standard -Testing of samples to be done at approved lab .				
h	The use of admixture, fly ash & crush sand should be as per concrete mix design.				
i	Note:- For lesser/Higer grades of concrete only cement difference shall be calculated based on RMC batch report/Design mix report and same shall be added/deducted including $\pm 15\%$ (Contractor's Profit) from the respective item rates.				
j	All the Concrete shall be measured in Cum basis only.				
k.	Rate shall inclusive of all heights, Sub structure, Superstructure, lift & leads, Scaffolding, shuttering, etc. Complete.				
III	CONCRETE WORK UPTO - SUBSTRUCTURE				
	PILE FOUNDATION				
7	Providing and Casting RCC bored cast in-situ Vertical piles as per IS 2911 (Part 1, Section 2) by boring through all kinds of soils/ Sand /Rock by hydraulic rotary rigs using temporary casing up to stable strata / bentonite mud circulation as specified, from tip to cut-off elevation of piles. Reinforced Cement Concrete work of filling the bore (after placement of reinforcement cage as per approved drawing) with M35 grade Ready Mix Concrete using 53 Grade Ordinary Portland Cement confirming to IS : 8112, of approved make and brand (ACC/ Ultratech/ JK) with minimum cement content of 400 kg/m ³ and with water -cement ratio, including the water contained in aggregates (10mm to max 20mm size), not exceeding 0.45, with approved plasticizer as specified, including placing of concrete from tip to minimum of 600mm above the specified cut-off level, breaking pile head to cut-off level and exposing pile reinforcement for embedment in pile cap, Disposing & levelling of bored/excavated material suitably at locations approved by the local authorities including all lead and lifts, all complete for piles having diameter of (Pile will be measured for payment for length between toe of the pile to cut-off level (bottom of pile cap) along the pile axis. (Reinforcement shall be paid separately). Cost of temporary casing shall be incurred by the Contractor.				

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
	Boring or chiselling through any obstacles, boulders, weathered rock, etc. upto founding level of piles for seating resting or socketing of piles. The rate quoted shall include demobilising the site with all plants, equipment, piling rigs, necessary personnel brought for piling work, including removal and transportation of debris away as directed, cleaning and reinstating site to original condition as acceptable by the Architect on completion of piling work.				
a	750mm Dia Pile	Rmt	2,430	10,400	2,52,72,000
b	600mm Dia Pile	Rmt	488	9,100	44,40,800
c	500mm Dia Pile	Rmt	1,170	6,000	70,20,000
8	TESTING PILE				
	Vertical load testing of piles in accordance with IS 2911(Part IV) including installation of loading platform by Kentledge/Anchor piles method and preparation of pilehead or construction of test cap and dismantling of test cap after test etc. complete as per specification & the direction of Engineer in-charge. Note: 1. Initial and Routine Load Test shall not be carried out by Dynamic method of testing. Note: 2. Testing agency shall submit the design of loading platform for the approval of Engineer-in-charge All diameters - Safe vertical load capacity				
	750mm dia to 500 mm dia - Safe vertical load capacity	Nos	2	2,94,000	5,88,000
9	Carrying out Dynamic Initial vertical load test on Bored cast-in-situ pile as specified in ASTM D4945 for specified test loads (750mm dia to 500 mm dia). Includes all plant, equipment, machinery, instruments, measuring devices, platforms and Kent ledge pile caps etc., dismantling and carting away entire setup, excavation and dewatering etc., all complete	Nos	4	1,32,000	5,28,000
TOTAL OF SECTION - III					3,78,48,800
IV	PLAIN CEMENT CONCRETE				
	Providing, machine mixing and laying plain cement concrete (PCC)				
10	Providing, machine mixing and laying plain cement concrete (PCC) (using OPC) as specified in drawing, using 20 mm down graded nominal size well-graded aggregate; in foundations, for bases of foundations, walls, plinths, rafts, plinth beams, floor slabs, pits to make up level etc.; coping, sill concrete, including compacting, curing, required centring/ shuttering with its removal, dewatering, cleaning, preparing surfaces, junctions, etc.; complete at all leads, levels, lift and depths (Sub structure) as per the drawing and to the satisfaction of the SBI/APMC.				
10a	PCC 1:3:6 (1 Cement : 3 Sand : 6 Coarse aggregates) in foundations & plinth beams. & also in sub structure & Super structure for all heights/Depth & leads.	Cum	703	7,650	53,77,950
10b	PCC 1:2:4 using 20 mm and downgrade metal (Coarse Aggregate).	Cum	100	8,200	8,20,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
	Note:- All PCC Work payment shall be done only for plan area and not more than 150 mm Projection outside the structural member over the PCC.				
	All concrete should be Ready Mixed Concrete/Cast in Situ & all raw material used for concrete like Sand, Cement, Metal Plasticizer, Fly Ash etc. Should comply with the norms of Green Building & should be approved by Structural Engineer / SBI/APMC and Green Building Consultant/s. For all small concreting activities Contractors may used Site mix with proper mix design duly approved by Structural engineer/ SBI/APMC. But it should be in 1 bag mixer & not in half bag mixer with proper Weigh Volumetric batching with required Pherna / Boxes and that too after approval of design mix report by Structural engineer/SBI/APMC for particular concrete & no hand mix concrete is permitted.				
V	RCC WORKS UP TO PLINTH LEVEL				
11	Providing and laying in position ready mixed/Design Mix, various grade concrete for reinforced cement concrete work				
	Providing and laying in position ready mixed/Design Mix, various grade concrete for reinforced cement concrete work, using (OPC) cement of approved manufacturer with cement content as per approved design mix, and manufactured in fully automatic batching plant and transported to site of work in transit mixer for all leads, lifts & heights / Depths, having continuous agitated mixer, manufactured as per mix design for reinforced cement concrete work at all leads, levels and heights/depths, including pumping of R.M.C. from transit mixer to site of laying, including the cost of centring, shuttering, strutting, scaffolding staging, Bracing, finishing the exposed surfaces properly, de shuttering and hacking the surfaces as desired curing etc. complete And with including making Cut outs, Holes, Core cut, Providing Sleeves Rungs with no extra cost Etc. (excluding steel reinforcement which is paid separately), including cost of admixtures in recommended proportions as per IS : 9103 to accelerate / retard setting of concrete, improve workability without impairing strength and durability as per direction of the SBI/APMC.				
	(Rate quoted should be for all heights & leads for all RCC work) in the footings of columns, Columns, Shear walls, beams, walls, pardis, fins, patli, floor slabs, etc.				
	All shuttering/form work materials shall be materials shall be of Good Quality Waterproof Ply / Aluform/MS only as per site requirement & in combination, all supporting materials like props, spans, H frames, Chavi etc. shall be of MS Material as directed/approved by SBI/APMC.				
a	In all types of footings / Pile cap -M35	Cum	976	12,600	1,22,97,600
b	In columns /shear walls / Lift below - up to Plinth -M35	Cum	245	17,700	43,40,040
c	In Plinth beams -M35	Cum	301	14,700	44,24,700
	TOTAL OF SECTION - IV				2,72,60,290

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
V	RCC WORKS - IN SUPERSTRUCTURE				
12	Providing and laying in position ready mixed/Design Mix, various grade concrete for reinforced cement concrete work				
	Providing and laying in position ready mixed/Design Mix, various grade concrete for reinforced cement concrete work, using (OPC) cement of approved manufacturers content as per approved design mix, and manufactured in fully automatic batching plant and transported to site of work in transit mixer for all leads, lifts & heights / Depths, having continuous agitated mixer, manufactured as per mix design for reinforced cement concrete work at all leads, levels and heights/depths, including pumping of R.M.C. from transit mixer to site of laying, including the cost of centring, shuttering, strutting, scaffolding staging, Bracing, finishing the exposed surfaces properly, de shuttering and hacking the surfaces as desired curing etc. complete And with including making Cut outs, Holes, Core cut, Providing Sleeves Rungs with no extra cost Etc. (excluding steel reinforcement which is paid separately), including cost of admixtures in recommended proportions as per IS : 9103 to accelerate / retard setting of concrete, improve workability without impairing strength and durability as per direction of the SBI/APMC.				
	(Rate quoted should be for all heights & leads for all RCC work) in the footings of columns, Columns, Shear walls, beams, walls, pardis, fins, patli, floor slabs, etc.				
	All shuttering/form work materials shall be materials shall be of Good Quality Waterproof Ply / Aluform/MS only as per site requirement & in combination, all supporting materials like props, spans, H frames, Chavi etc. shall be of MS Material as directed/approved by SBI/APMC.				
a	Grade Slab -M20 at plinth level	Cum	405	12,500	50,62,500
b	In columns /shear walls- Including Cut-outs, Holes, Sleeves, Rungs Pergola At all levels Etc. - M35	Cum	1,344	23,900	3,21,21,600
c	Column - Grade M 40	Cum	10	24,300	2,43,000
d	In Floor beams at all levels/ Lintel beams/ Patli /Sill levels beams Etc. - M35	Cum	3,130	23,900	7,48,07,000
e	In slabs at all levels - M35	Cum	4,050	20,200	8,18,10,000
	RCC walls /Retaining walls/ Fins/ Parapet wall/ Drop Pardies/ up stands for staircases Pardis/ Walls of UGWT, OHWT. Including Cut-outs, Holes, Sleeves, Rungs Pergola At all levels Etc.-M35				
f	RCC Wall / Lift Wall -M35	Cum	670	22,700	1,52,09,000
g	RCC Parapet Wall -M35	Cum	70	28,800	20,16,000
h	In sloping slabs and steps for staircase -M35	Cum	290	20,200	58,58,000
i	Chajja Concrete -M35	Cum	60	20,200	12,12,000
	TOTAL OF SECTION - IV				21,83,39,100

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
13	REINFORCEMENT STEEL WORKS				
	Providing & laying including straightening, cutting, bending and binding in position of steel reinforcement bars of approved manufacturers confirming to BIS, to correct shape and size as per detailed drawings, specifications and approved bar-bending schedules, Including preparing & Maintaining bar bending schedules in substructure / superstructure walls, columns, Shear Wall, slabs, beams, retaining Wall, sloping slab & steps for staircase, chajja, lintel, patali, Fins, all types of Paradi, pergola, Including water tanks, Septic tank etc.; with correct size Super fine micro concrete cover blocks and annealed binding wire (G.I.) of 18 gauge including dewatering to keep site dry etc. complete at all heights and leads to the satisfaction of the SBI/APMC. Measurement of steel is strictly as per I.S. Code. No extra for bending will be permitted.				
	With TMT / 550 D conforming to IS : 1786/1139 High yield strength corrosion resistant bars with respect to physical properties and of approved chemical properties in all RCC work (epoxy bonded powder coated steel to be considered) rates are inclusive of all heights & leads.				
	Weights shall be calculated based on mode of measurement as specified in Technical Bid.				
a	SUB STRUCTURE	MT	439	1,36,000	5,97,04,000
b	SUPER STRUCTURE	MT	1,205	1,36,000	16,38,80,000
	TOTAL OF SECTION - V				22,35,84,000
VI	MASONRY WORK				
	Measurement shall be considered Per Cum or Per Sq m for respective items deduction shall be done for RCC Items as per relevant BIS/ standards				
	Rate shall inclusive of all heights, Sub structure, Superstructure, lift & leads, Scaffolding, etc. Complete.				
14	Burnt Brick Masonry in Foundation				
	Providing and laying Brick Wall with well burnt uniform size and colour locally available approved quality bricks with 230 mm thick and jointing them together with sand cement joints 1:4 of thickness not less than 12 mm thk, with minimum crushing strength not less than 50kg/cm2 including curing, scaffolding, Lifts ,raking the joints at all levels complete				
a	Foundation - CM 1:4	Cum	382	8,800	33,61,600
b	Super Structure - CM 1:5- External walls	Cum	2,450	10,400	2,54,80,000
15	CONCRETE SOLID BLOCK				
	Providing and constructing 200mm thick Solid Concrete Block work in walls, piers, fins and architectural features at all levels using approved blocks of size 400x200x200mm and of minimum crushing strength 35kg/sqcm in cement mortar 1:5 including curing, scaffolding, staging, raking out joints, leads and lifts, etc., complete.				
	All Floors and all heights	Cum	100	12,000	12,00,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
16	200 mm thick Autoclaved Aerated Cement block masonry Providing and laying Autoclaved Aerated Cement blocks masonry in structure at all floor level, with 200mm thick AAC blocks of approved manufacturers, having rough surface to receive the plaster effectively in Sub Structure/ Super structure above plinth level in all level including RCC band/ Patli/ Beam with required form work , Steel, Concrete Etc. at sill level and lintel level or wherever required with approved block laying polymer modified adhesive mortar with 1:4 all complete as per direction of SBI/APMC. AAC Blocks shall have minimum density in oven dry condition of 650Kg/M3 and Minimum Compressive strength of 5 N/mm2 as per IS 2185 Part III, to correct line, level and plumb including raking out joints, curing, scaffolding etc at all heights & leads as per detailed drawing. complete.				
	All Floors and all heights	Cum	2,520	9,200	2,31,84,000
17	100 mm thick Autoclaved Aerated Cement block masonry Providing and constructing 100 mm thick AAC block masonry in structure at all floor level, Having rough surface to receive the plaster effectively, blocks conforming to IS 2185 (Part-III) masonry of approved manufacturers with approved block laying polymer modified adhesive mortar with 1:4 per manufacturer's instructions, including providing RCC patli beam(Lintel) of 100mm thick at every 1.00m c/c with 2 nos. 8 mm dia Tor bars and 6 mm dia links at 300 c/c in cement concrete of mix M 20, required formwork complete with raking out joints, curing, independent double legged scaffolding as per specifications etc. at heights, depths and leads as directed by PM to his entire satisfaction. (Patli beam/ Lintel RCC & Steel will be paid seperately).				
	All Floors and all heights	Sqm	5,050	1,200	60,60,000
	TOTAL OF SECTION - VI				5,92,85,600
VII	FLOORING WORK				
	Note:- All Flooring/dado /cladding works: Measurements shall be the exposed, visible finished surface area of application. No extra/separate payment for Groove , chamfer edge Polish, moulding, rounding, nosing, overlapping portions, chiseling in walls, tile, etc.				
	Rate shall inclusive of all heights, Sub structure, Superstructure, lift & leads, etc. Complete.				
18	VITRIFIED TILE FLOORING - DOUBLE CHARGED Providing and laying in-position Double charge Vitrified tiles flooring including Skirting of thickness (10mm +/- 1mm) and approved make and colour conforming to IS :15622, rate including fixing the same over average 20mm thick CM 1:6 bedding using approved quality tile adhesive of approved make, uniformly in line and level in approved design and filling the joints with epoxy based joint filling compound of matching shade curing the joints, etc. as per instruction of SBI/APMC complete.				
	Double charge polished gloss/ matt finished Vitrified tiles in flooring/ Skirting of size 800 x 800 mm / as approved by the SBI/APMC (Basic rate Rs. 100/- per sqft)	Sqm	12,330	2,100	2,58,93,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
19	VITRIFIED FLOORING FOR TOILETS - (MATT FINISH)				
	Providing & laying in-position Vitrified tiles in floor of approved make and colour uniformly in line, level and in approved design, including laying avg. 20 mm thick bed of CM 1:4 (1 cement : 4 coarse sand) and filling the joints with epoxy based joint filling compound of matching shade, curing the joints etc complete with waterproof compound.Rates to include as per drawing flushed with finished surface of wall in line and level and 6 mm wide groove to be made in wall at the junction of skirting and finished surface of wall etc.complete matching to flooring as per direction of Engineer-in-Charge. Matt finished Vitrified tiles: Size of Tile 600 x 600 mm x 10mm (+/- 1mm)				
	Vitrified tiles of size 600 x 600 mm (Toilets flooring) (Basic rate Rs. 66/- per sqft)	Sqm	1,480	1,700	25,16,000
20	CERAMIC TILES FOR TOILET AND KITCHEN WALL DADO				
	Providing and fixing in position premium quality joint free /paper cut edges digital print ceramic wall tiles of size 300 X 600 mm in walls/dado conforming to IS: 15622 (thickness to be specified by the manufacturer) of approved make, colours and shades as per sample approved by Engineer in-Charge fixed over 15mm thick rough finished backing plaster in CM 1:4 fixed with neat cement slurry including filling joints using white cement joint filling compound with matching pigment including acid cleaning of tiles complete in all respect and at all heights, levels and locations. Rate includes cost of CM 1:4 backing plaster with waterproofing compound and PVC beading at all exposed edges or corners and execution of work all heights and locations including necessary scaffolding etc. complete.				
	Ceramic Digital Wall Tiles of size 300 x 600 mm / as approved by the SBI/APMC (Toilets, wash basin, Kitchens, Utility area dado) (Basic rate Rs. 60/- per sqft)	Sqm	6,960	1,550	1,07,88,000
21	VITRIFIED TILE FLOOR WOODEN PATTERN - BALCONY				
	Providing & laying in-position vitrified tile flooring of wooden pattern and skirting of approved make and colour uniformly in line, level and in approved design, including laying avg. 20 to 25 mm thick bed of CM 1:4 (1 cement : 4 coarse sand) and filling the joints with epoxy based joint filling compound of matching shade, curing the joints etc complete. Rates to include providing & fixing 100 mm high skirting as per drawing flushed with finished surface of wall in line and level and 6 mm wide groove to be made in wall at the junction of skirting and finished surface of wall etc.complete matching to flooring as per direction of Engineer-in-Charge.				
	Matt finished Vitrified tiles: Size of Tile 1200 x 200 mm x 10mm (+/- 1mm) (Basic rate Rs. 75/- per sqft)	Sqm	1,730	1,800	31,14,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
22	PODIUM - HEAVY DUTY EXTERIOR GRADE VITRIFIED TILE				
	Providing & laying in-position Vitrified exterior grade heavy duty tile flooring and skirting of approved make and colour uniformly in line, level and in approved design, including laying avg. 20 to 25 mm thick bed of CM 1:4 (1 cement : 4 coarse sand) and filling the joints with epoxy based joint filling compound of matching shade, curing the joints etc complete. Rates to include providing & fixing 100 mm high skirting as per drawing flushed with finished surface of wall in line and level and 6 mm wide groove to be made in wall at the junction of skirting and finished surface of wall etc.complete matching to flooring as per direction of Engineer-in-Charge.				
	Matt finished Vitrified tiles: Size of Tile 600 x 600 mm x 10mm (+/- 1mm) (Basic rate Rs. 80/- per sqft)	Sqm	1,000	1,900	19,00,000
23	TERRACE HEAT RESITANCE TILE				
	Providing and fixing in position Heat Resistant Terrace Tiles Flooring and skirting using tiles of size 400 mm x 400 mm x 10 mm of approved makes and shade with SRI (solar refractive index) > 78, solar reflection > 0.70 and initial emittance > 0.75 on waterproof and sloped surface of terrace, laid over avg 20 to 25 mm thick C.M. 1:6 (1 cement : 6 coarse sand) bedding and grouting the joints with mix of white cement & marble powder in ratio of 1:1, including machine cutting, grinding and polishing to achieve smooth surface complete in all aspects as specified and directed by the Engineer/Architect. (Basic rate Rs. 50/- per sqft)	Sqm	2,080	1,900	39,52,000
24	WHITE CERAMIC TILE - WATER TANK				
	Providing and laying in position Ceramic White tile of size 300mm x 300mm of approved make and shade in proper slope, line, level with neat Cement slurry on inside walls and floors of Under Ground and Over Head Water Tanks etc. complete. (Basic Rate for ceramic tiles - Rs. 46/- per Sft)	Sqm	880	1,500	13,20,000
25	KOTA STONE IN STAIRCASE AREA				
	Providing & laying machine cut, machine polished Kota stone in floor landing & Midlanding / skirting / treads / risers / sills etc. in Risers of steps, 18-20, mm thk, in single piece up to 1600 mm length over CM 1:3 bedding 15 mm thk. (Average) as per drawings, details, and instructions of SBI/APMC etc. including providing & laying bedding mortar, grouting the joints with cement slurry, mirror polishing, cleaning etc. complete as per specification and as directed at all locations and heights, etc. Complete. (Staircase Area) (Basic rate Rs. 40/- per sqft)	Sqm	1,080	2,600	28,08,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
26	GRANITE KITCHEN PLATFORM				
	Kitchen Counter Top : Providing & fixing in position Kitchen platform/Vanity Counters with minum 600mm width and height at 750 mm height Vertical supports or as per Architectural design: Made out of 18 mm thick white marble slabs sandwich supports (two slabs joined together using Araldite adhesive to full surfae area) placed at specified spacing as per detailed drawing and suitable to house modular kitchen under the counter Trolleys. Rate includes providing vertical garnite band to vertical supports with moulded/ polished edges. Kitchen Counter Top : Finished with 18+/-2mm approved qulaity, Colour, shade and textutre mirror polished Granite slab (minimum length of slab shall not be less than 1.80 Meter for Kitchen counter and one single piece for Vanity Counters for wash basin etc.) resting on 18mm thick white marble base support slab (in one piece) and fixed with CM 1:2 or approved tile adhesive as directed at site. Front Facia: 50 to 75mm wide Granite band in one piece not less than 1.80 RM in length for kitchen counter and required length in one piece for Vanity counters with full round moulded mirror polish edges complete as per Architectural working drawing. Making hole/sleeves for Gas pipe etc complete.	Sqm	380	4,900	18,62,000
	Rate should include chasing walls, floors to embed vertical supports and counter top using CM 1:2 including making cutout for kitchen sink, under the counter wash basin, core cutting for pillar cock, bottle trap etc. complete as per directions & instructions of the Architect and Detail Drawings. (Basic rate Rs. 180/- per sqft)				
27	GRANITE COUNTER FOR WASH BASINS				
	Providing & Fixing Pre polished Granite of thickness 16 to 18 mm in Oval/Rectangular/Square shape for over / under counters to wash basin on to the wall in cement mortar 1:3 including making cutout for basin, fixing, the front & side fascia of 100 mm width shall be provided of Pre polished Granite, also fixing of Horizontal / vertical kadappa 25 to 30mm thick for support as per design & dwg. Item includes cutting, polishing, nosing, chamfering, edge polishing, rounding, groove making , Curing, finishing joints with matching pigment, curing etc complete as directed by the SBI/APMC . (Top Plan Area Shall Be Measured for Payment) (Basic rate Rs. 215/- per sqft)	Sqm	180	5,300	9,54,000
28	GRANITE STONE CLADDING - Lift facia				
	Providing & laying 18-20 mm thick polished approved Granite Stone in cladding work over 15 mm thk CM 1:3 backing, fixed with approved quality Stone adhesive of approved make, filling with matching cement slurry, curing, epoxy grouting, rounding corners, cleaning etc. Complete. (Lift Cladding at all floor). as per design and Drawing given by SBI / APMC. (Basic rate Rs. 210/- per sqft)				
	All Floors	Sqm	940	5,200	48,88,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
29	GRANITE FLOORING IN CORRIDOR				
	Providing and laying in position 18mm ± 2mm thick machine cut pre-polished Granite stone flooring and skirting in approved colour, shade, design and patterns as per architectural drawings ilaid over 20 mm (average) thick cement mortar 1:3 (1 cement : 3 coarse sand) bedding fixed and jointed with neat cement cement slurry and filling joints using white cement slurry admixed with pigment of matching shade or epoxy based joint filling compound of approved make including grinding, mirror polsihing, curing, nosing, making grooves etc. complete as specified in the Architectural drawings and as directed by the Engineer-in-Charge. (Basic rate Rs. 215/- per sqft)	Sqm	3,600	5,200	1,87,20,000
30	ITALIAN MARBLE FOR ENTRANCE LIFT LOBBY- FLOORING/ DADO				
	Providing and fixing in position 18 mm ±2mm thick Pre-polished Italian marble flooring using of uniform and homogeneous colour, shade and texture as per approved pattern and design including Inlay pattern as per Architectural drawings and sample approved by the Bank with machine cut edges, laid over average 40 mm thick CM 1:4 (1 cement : 4 coarse sand) bedding fixed with matching neat white cement slurry or approved tile adhesive of Roff/Bal Endura/Fevimate or equivalent approved by the Bank laid to level as required, curing etc. complete. The rate includes machine polishing of floor in mirror finished as per sample aproved by the Bank and as directed by the Engineer-in-charge. (Basic Rate of Italian Marble : Rs.600/- per sft)				
	In Flooring using Italian Marble slabs not less than 1.50 Sqm in plan area (execpt on edges/ends).	Sqm	200	11,000	22,00,000
31	GRANO FLOOR				
	52 mm thick cement concrete flooring with concrete hardener topping, under layer 40 mm thick cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) and top layer 12 mm thick cement hardener consisting of mix 1:2 (1 cement hardener mix :2 graded stone aggregate 6 mm nominal size) by volume, hardening compound mixed @ 2 litre per 50 kg of cement or as per manufacturer's specifications. This includes cost of cement slurry, but excluding the cost of nosing of steps etc. complete.				
a	Stilt Floor Parking Area	Sqm	3,000	1,100	33,00,000
b	First floor Parking Area	Sqm	3,020	1,100	33,22,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
32	GRANITE STONE FRAMES FOR TOILETS				
	Providing & fixing 18mm thk Granite (Jet Black) gang saw cut, mirror polished, premoulded and prepolished, machine cut in 2 levels of sills and jambs for windows and ventilators, Doors, slabs of approved quality and colour with machine polished, to cover the wall thickness, in proper line / level, plumb, with edge rounded or chamfered over a CM 1:3 bedding including neat cement float / tile adhesive (between two stones), filling joints with cement slurry to match the colour of the slabs, curing polishing, joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing, moulding and polishing to edges to give high gloss finish cleaning etc complete. as per the Drawings details and instructions of SBI/APMC etc Complete. (Window sills & jam, sliding door, Toilet / Utility area Door Frame) (Basic rate Rs. 215/- per sqft). Plan area in sq.m will be measured.	Sqm	192	9,900	19,00,800
	TOTAL OF SECTION - VII				8,94,37,800
VIII	JOINERY				
	Note:- Before fixing Door frame back side of door frame shall be treated with anti termite/black japan paint in 2 coats. Also Frame shall be fixed with minimum 3 Holdfast on each side and shall be fixed in concrete which is to be done in wall to take hold fast				
	Door Area including Frame shall be measured above finished floor level for the Payment.				
	Rates shall be inclusive of hardware items of approved makes.				
33	DOUBLE & SINGLE SHUTTER WOODEN FRAMES				
	Providing and fixing in position approved quality second class seasoned Teak wood door frame in design, shape and profile size as per drawing with black japan paint to the surface in contact with plaster/ masonry/ RCC. The over all sectional size as required including fixing in position with 6 nos. M.S. holdfasts anchored in 1:2:4 cement concrete of approximate size 200 x 200 x 150mm, making arrangements for fixing aldrop, tower bolts, slip way (tadi) in the frame. The frame is finished with two or more coat of melamine polish after preparing surface complete. Rates to include necessary hardware, screws, etc. Complete. (For Main Door and Internal Doors)	Cum	40	3,60,000	1,44,00,000
34	DOUBLE SHUTTER DOORS				
	Providing and fixing in position 35 mm thick Laminated Flush Door Shutters. Rates to include necessary SS hardware like 4 nos. heavy duty 125 mm Butt Hinges, 300x8mm Tower bolt, 2 nos. 200mm long 10mm dia door handle, 300 x 12 mm Al Drop with Locking Arrangement, 300 X 12 mm Door Latch without Lock Hole Bracket, door buffers with magnetic cmainatcher, hanging type door stopper etc. Complete. approved by the SBI/APMC. The rates are inclusive of all the hardware items specified. All fittings should be SS make (brush/ glossy finish).	Sqm	25	11,900	2,97,500

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
35	SINGLE SHUTTER - Main Door Providing and fixing in position 35 mm thick Decorative Laminated Flush Door Shutters. Rates to include necessary SS hardware like 4 nos. heavy duty 125 mm Butt Hinges, 300x8mm Tower bolt, 2 nos. 200 - 300mm long 10mm dia door handle, 300 x 12 mm Al Drop with Locking Arrangement, Mortise lock with handle (Basic Price 2500/- each) and computerized key, door buffers with magnetic catcher, door stopper, magic eye lens etc. complete. approved by the SBI/APMC. The rates are inclusive of all the hardware items specified. Also 12 mm thick ply and 1 mm laminate boxing around the door frames on both sides for 150 mm wide as per the detail drawing and dwsign approved by SBI/APMC. All fittings should be SS make (brush/ glossy finish).	Sqm	295	10,700	31,56,500
36	SINGLE SHUTTER - Internal Doors Providing and fixing in position 35 mm thick Laminated Flush Door Shutters. Rates to include necessary SS hardware like 4 nos. heavy duty 125 mm Butt Hinges, 300x8mm Tower bolt, 200 x 10 mm , Mortise lock with 300mm handles on both sides, door buffers with magnetic catcher, door stopper etc. Complete. approved by the SBI/APMC. The rates are inclusive of all the hardware items specified. All fittings should be SS make (brush/ glossy finish).	Sqm	1,300	6,300	81,90,000
37	TOILET SHUTTER Providing and fixing in position 30mm thick approved quality and design water proof, solid FRP Doors shutters as per sample approved by the SBI/APMC for wash rooms, Utility area. Each door shutter will be provided with 125X30 mm 3 Nos. butt hinges, SS tower bolts, SS handle 150 mm, with knob type lock etc. Complete. The rates are inclusive of all the hardware items specified. All hardware shall be of approved makes.	Sqm	540	4,600	24,84,000
38	2 Hr Fire Rated Door & shaft Doors Providing and fixing 2- Hours fire resistant door (IS 5509 or CBRI Fire certified) with Vision panel of size 300 X 300mm, Panic Bar, Shutter and other necessary approved fittings/ hardware with Manufacturers certificate as per National Building Code 2016 guidelines etc. complete & as per the drawing by SBI/APMC. a.) Frame of size 143x58 mm and thickness of 1.6 mm thick skin pass galvanized iron sheet (Conform to IS 277) formed to single or double rebate profile b.) Shutter of 46 mm thickness and constructed from 1.2 mm thick skin pas galvanized iron sheet (Conform to IS 277) infill Rockwool / Mineral wool/ Honey comb craft paper c.) Vision glass of 5mm thick clear/wired glass Circular or Rectangular d.) Finish : Primed with zinc phosphate stoving primer and finished with Polyurethane aliphatic grade or epoxy paint as required of thickness 60 micron ed.) Accessories : Min 8 Anchor fastener of HILTI make, min 4 SS Ball Bearing Butt Hinges, Mortise sash lock with lever handle, Mortise dead bolt , Mortise latch , Door closer , Panic bar and external trim lock and Flush bolts complete. The rates are inclusive of all the hardware items specified.				
38 a	Staircase Door	Sqm	160	21,200	33,92,000
38 b	Shaft Door	Sqm	80	21,200	16,96,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
39	SINGLE GLAZED TOUGHENED GLASS PARTITIONS				
	Slim line glass partitions:- Providing and fixing 12mm thick straight toughened glass full height partitions comprising as under: (i) Framing:- Made using 80 x 18mm, approved make, powder coated aluminium sections of SLIM LINE series at floor and false ceiling level as per manufacturers' specifications. (ii) Door Frames/Openings: Made using 105x25mm, approved make, powder coated aluminium WIDE LINE sections for Door Frame. The two vertical sections of the frame shall be extending upto false ceiling level and horizontal section shall be provided at door top level. The rate shall be inclusive of all fittings and accessories viz. stake1, stake3 and wedge 3G/B from FFL to the bottom of false ceiling height.				-
	(iii) Glazing:- Providing 12mm thick toughened clear glass panels of approved make to the frame sections fixed properly as per manufacturers' specifications including sealing glass joints using approved silicon sealant on both side. (iv) Fixing flush stop of size 20mmx15mm for door frame visible joint, grooves of size 12 mm etc complete. Rate shall include for providing transport charges, handling, loading and unloading & installation of all necessary hardware's, scaffolding, wastage, providing necessary structural supportsto the framework with walls/roof ceiling etc. complete all as per manufacturers specifications as directed.	Sqm	30	9,400	2,82,000
					-
40	GLASS DOOR				-
	Fully glazed door: Providing and fixing in position frameless glass door shutter made using 12mm thick toughened glass fixed to the partition (glazed / solid / gypsum partitions) using heavy duty floor spring with pivot of approved make including necessary cutting of floor and making the surface good to its original position including all necessary Patch fittings of approved make and fixtures & hardwares. Approved make: DORMA Top Pivot - PT 24, Top Patch Fittings - PT 20, Bottom Patch fittings - PT 10, Floor Spring - BTS 75 V, Corner lock with strike plate and Euro Profile Cylinder - US 10, Handle - TG DIH600 - 600mm length) or equivalent brand fittings from Hettich / Hafele / as per sample approved by the Bank. Mode of Measurement: Door size shall be measured	Sqm	25	26,000	6,50,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
41	UPVC WINDOWS & DOORS				
	Providing & fixing in position factory made uPVC white colour casement /Sliding /casement cum fixed fully glazed Doors/ sliding window upto 1.50 M to 2.4M in height dimension comprising of 6mm thick Toughened glass for all window & 3.5mm thick frosted glass for ventilator, uPVC multi-chambered frame with in-built roller track and sash extruded profiles duly reinforced with with 15. to 2.52 mm thick depending upon the window glavanised mild steel section made from roll formaing process of required length (shape & size according to uPvc profile), appropriate dimensions of uPVC extruded glazing beads and uPVC extruded interlocks, EPDM gaskets, stainless steel (SS 304 grade) friction hinges, zinc alloy (white powder coated) casement handles, wool pile, touch locks with hooks, with single nylon rollers (weight bearing capacity to be 80/ 150 Kg), SS fasteners 100 x 8 mm size for fixing frame to finished wall and 5 degree gradient slope in track (drainage track) comes with threshold, low threshold without threshold option & necessary stainless steel screws etc 8 mm thick toughened glass panes (ST 136), fixed in shutters, flush handle, C handle, D handle with locking arrangements as per manufacturers specifications etc. Complete.				
	Profile of frame & sash shall be mitred cut and fusion welded at all corners, mullion (if required) shall be also fusion welded including drilling of holes for fixing hardware's and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealant of approved quality, including Mosquito net all complete as per approved drawing & direction of SBI/APMC. Variation in profile dimension in higher side shall be accepted but no extra payment on this account shall be made.				
41a	UPVC Window sliding type - 6 mm thk toughened glass (Two/Three Track system)	Sqm	1,070	21,800	2,33,26,000
41b	UPVC Window Top Hung type - 6 mm thk toughened glass (Top hung)	Sqm	60	21,800	13,08,000
41c	UPVC French Door - 8 mm thk toughened glass	Sqm	700	19,000	1,33,00,000
41d	UPVC Ventilators - 3.5 mm thk frosted glass	Sqm	130	18,700	24,31,000
	TOTAL OF SECTION - VIII				7,49,13,000
IX	STEEL WORKS				
42	MS Grills				
	Providing and fixing M.S. grills of required pattern in frames of windows etc. with M.S. flats, square or round bars with all welding cutting, fitting, finishing etc. including priming coat with approved steel primer, two or more coats of enamel paint & as per approved design/drawing all complete as directed by the SBI/APMC (window , well & water tank).	Kg	25,310	300	75,93,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
43	STRUCTURAL STEEL WORKS - IN COMPOUND WALL				
	Providing and fabricating structural steel work in rolled sections like joists, channels, angles, tees etc. as per detailed design and drawings or as directed including cutting, fabricating, hoisting, erecting, fixing in position making riveted / bolted /welded connections without connecting plates, braces etc. and including one coat of anti corrosive primer and over it two or more coats / up to satisfaction of SBI/APMC of oil painting of approved quality and shade, inclusive all heights, Sub structure, Superstructure & leads, etc. Complete.				
	Theoretical weights shall be calculated for all structural items but excludes weights of Nut Bolts and accessories				
	For Compound wall and Gate	Kg	10,400	170	17,68,000
44	SS RAILING WITH TOUGHENED GLASS - BALCONY				
	Providing fabricating & Fixing 1200mm high / as per instructed by SBI/APMC Glossy/mat finish S.S. 304 Grade Stainless steel glass railing with 50mm diameter and 1.5mmthick at top,40mm diameter and 1.5mm thick vertical supports spaced at 1.2m centre to centre and 12mm thick toughened glass ncluding fabricating , erecting , necessary welding , grinding , finishing, buffing to stainless steel pipe fixtures, rate also includes of all necessary Patch fittings, etc. Complete as directed and as per the design and drawing given by SBI/APMC .				
	Using Stainless Steel Grade 304 Sections+12 mm thkToughened glass for railing - Balcony	Rmt	1,690	9,600	1,62,24,000
45	STAINLESS STEEL GRILLS / RAILINGS				
	Providing & fixing Stainless Steel 304 Grade Railing over stair case steps / landings at 1050 mm height from FFL comprising of 75 mm dia and 16 gauge SS top glossy finish pipe supported by 50 mm dia, 10 gauge SS glossy finish at 600 mm c/c (in alternative treads in stairs), 2 - 4 nos 25mm dia horizontal SS pipe in between with proper fixing arrangements including providing SS round cover at a junction of each support and hand rails, rate also includes of all necessary Patch fittings etc all complete as per specification and the pattern as provided in the drawing and as per direction of the SBI/APMC.	RM	600	3,500	21,00,000
46	MOTORZED ROLLING SHUTTER:				
	Rolling shutters with machine rolled lath sections from 18G (1.22MM) bright CRCA FLAT strips in one piece without any welded joints rolling centres of not more than 76mm interlock with each other ends locked with heavy type M.S. Clips an effective depth of 12mm complete with reinforced 10G(3.15) lock plate with 35x35x5mm angles, 10G (3.15)pressed Guide channels and 6 mm thick bracket plates. The shaft heavy duty M S Pipes without springs (JIELONG) or (AUTOZON) side Motor, 6mm side bracket plates for the automation including push button, control switch, in case of power failure we can have the option of manual override, The hood cover made out of 20G (0.90)sheet with 25 x 3 mm angle stiffeners with one coat of red oxide primerand with two 2 or more coat of synthetic enamel paint. (Outer dimensions including hood will be measured for payment purpose.)				
	Rolling Shutter - 3 x 4	Sqm	20	7,200	1,44,000
	TOTAL OF SECTION - IX				2,78,29,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
X	PLASTERING WORK				
	Note:- All plaster works Measurements shall be considered only as Surface Areas of application and no extra/separate payment for edges, corners, groove, strips, bands etc.				
	Only MS Scaffolds shall be used for the all Plastering Works				
	Rate shall inclusive of all heights, Sub structure, Superstructure, lift & leads, scaffolding etc. Complete.				
	INTERNAL CEILING PLASTER				
47	Providing Internal cement plaster for ceiling about 6 mm average thickness in a single coat in cement mortar 1:4 with Gujrat sand at all heights & leads to concrete surface in all positions including necessary scaffolding (Single/ Double), lifts racking out joints, curing complete and hacking with light hammer to all RCC surfaces complete.				
	All Floors and all heights	Sqm	27,810	320	88,99,200
48	INTERNAL CEMENT PLASTER TO ALL INTERNAL SURFACE (BRICK , BLOCK & RCC SURFACES)				
	Providing internal cement plaster about 12 mm avg. thickness in a single coat in cement mortar 1:4 without neeru finish with Gujrat sand to concrete or brick surface in all positions, heights & leads including scaffolding(Single/ Double.), curing, etc. complete. Including Providing fibre-mesh at joints of brickwork and concrete and also while making good the chase cut for the purpose of electrification / Plumbing , Filling all gapes along the wooden door frames, Making moist the surface before plastering & Curing the plaster surface for 15 Days. Rate includes adding polypropylene fibre (Recron - 3S Reliance IL or approved equivalent) of approved quality as per proportion specified in the manufacturers specifications complete as directed by the SBI/APMC.				
	All Floors and all heights	Sqm	50,020	410	2,05,08,200
49	EXTERNAL CEMENT PLASTER TO ALL EXTERNAL SURFACE (BRICK , BLOCK & RCC SURFACES)				
	Providing and laying external Cement Plaster, average 20 mm thk, in two coats, at all heights with first coat of av. 12 mm thk in Cement Mortar 1:5 using good quality Gujarat sand, and second coat of average 6 mm thickness in C.M. 1:4, including necessary staggering, scaffolding, curing, roughening the first coat to receive second coat of plaster etc, providing approved water proofing compound @1 kg per bag of cement or as specified by the manufacturer and adding polypropylene fibre of approved make as per manufacturers specifications to prevent shrinkage cracks, complete as directed by the Engineer-in-charge.				
		Sqm	13,157	600	78,94,200
TOTAL OF SECTION - X					3,73,01,600

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
XI	PAINTING WORKS				
	Only MS Scaffolds shall be used for the all Works				
	Rate shall inclusive of all heights, Sub structure, Superstructure, lift & leads, scaffolding etc. Complete.				
50	Providing and applying 2 or more coats of Birla putty on internal walls of average thickness of 3 to 4 mm in line & level including its rubbing with sand paper to ensure smooth & even surface including necessary scaffolding etc. complete.	Sqm	500	180	90,000
51	INTERNAL PAINT				
	Preparing the earlier treated surface by scraping rubbing with sand paper applying two coats of 3 to 4 mm Birla white putty including its rubbing with sand paper to ensure smooth surface including applying one coat of approved make primer and 2 or more coats of approved colour and shade of Acrylic Emulsion Paint with low VOC content to internal walls/ceiling of the flats including Lobby areas, Staircase, common areas etc. including necessary scaffolding etc. complete.				
	All Floors and all heights	Sqm	77,830	440	3,42,45,200
52	Exterior Grade Acrylic Emulsion Paint with low VOC content				
	Providing and applying Acrylic Emulsion with low VOC content / ACE of approved make or approved equivalent, colour, shade & quality to new surfaces in two or more coats over a coat of primer, filling up the services and over plaster surface including scaffolding all complete at stilts and podiums as directed.				
	Note: Rates to include cleaning afterwards, necessary scaffolding / Cradle, debris safety netting, labour etc complete.				
	All Floors and all heights	Sqm	770	300	2,31,000
53	EXTERNAL PAINTING WITH HIGHER GRADED ELASTOMERIC PAINT & TEXTURE PAINTING				
	a) Preparing the existing surface by scrapping and rubbing with sandpaper to remove all loose adhered particles followed by washing with water using wire brush to ensure thorough cleaning. Providing & applying rough tough putty over the prepared surface with agreed designed by SBI/APMC .				
	b) Providing and applying Texture plaster with finishing with texture material of approved make in 3 to 4 millimeter thickness on previously plastered surface, including Plaster Groove 6mm thickness or Tape Grooves 35 to 45 mm thickness or as required, in all position including preparing the surface, scaffolding etc. complete.				
	c) Providing and applying one coat of water based acrylic anti fungal Primer of approved quality and make as per sample approved.				
	d) Providing and applying one coat of fibre reinforced membrane paint water proofing Coat of "Protek Basecoat" of Asian/ "Weather coat Roof Guard" of Berger / "Aqua Touch" of ICI or equivalent approved by the SBI having crack bridging ability up to 2.00 mm, superior anti-carbonation & water proofing property.				

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
	e) Providing and applying 2 or more coats of approved quality External Acrylic Emulsion paint, Exterior grade having anti fungal property, resistance to dust & dirt accumulation, resistance to fading of colours etc. of approved colour and shade Apex Ultima (Asian) / Weathercoat All Guard (Berger) / Weather Shield Max (ICI) or equivalent approved by the SBI.				
	Note: Rates including 10 years warranty by the manufacturer in the prescribed format after cleaning , scaffolding/Cradle, debris safety netting, labour etc complete.				
	Note:- All Painting Works shall be Measured as per application surface area no extra/separate payment for Band, grooves, strips painting. For Grills /Jali MS Gates if not mentioned specifically /separately. Payment area shall be calculated as per Mode of Measurement given in Technical Bid Document/ BIS/As per Standards	Sqm	10,070	550	55,38,500
54	OIL BOUND DISTEMPER				
	Preparing earlier treated surface by scraping, rubbing with sand paper, applying two coats of Birla white putty, wherever required, to level the surface, and its rubbing with sand paper, to ensure smooth surface, suitable to receive paint and then applying 2 or more coats of approved colour and shade of Oil bound distemper with low VOC content over a coat of approved primer complete as per manufacturers' specifications to internal walls/ceiling including necessary scaffolding etc. complete.	Sqm	100	120	12,000
55	SYNTHETIC ENAMEL PAINT				
	Providing and applying Synthetic Enamel Paint on doors / windows frame, shutters, railing, grills, gates etc. in two or more coat of approved brand and shade including a priming coat as per manufacturers' specifications, after necessary preparation of surface with scrapping, sand paper rubbing, applying putty, including necessary scaffolding etc.complete as directed by the Engineer-in-charge.	Sqm	280	180	50,400
	TOTAL OF SECTION - XI				4,01,67,100
XII	WATER PROOFING WORKS				
56	All the waterproofing items shall be carried out by the reputed approved agency as mentioned in the tender and duly approved by the SBI/APMC. Item includes sealing of pipes/sleeve at no extra cost. The treatments shall be tested for water tightness and shall be guaranteed for water tightness in bone dry conditions for at least 10 years period during which period if any leakage is observed, the same shall be rectified to the satisfaction of the Owners. Such undertakings shall be furnished on Rs. 500/- stamp to cover all the above conditions.				

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
a	Waterproofing to Terraces.				
	Providing and laying in-position integral cement based water proofing treatment including preparation of surface as required for treatment of roofs, balconies, terraces, top terraces etc consisting of following operations:				
	a) Applying a slurry coat of neat cement using 2.75 kg/Sq m. of cement admixed with water proofing compound conforming to IS. 2645 and approved by SBI/APMC over the RCC slab including adjoining walls upto 300 mm height including cleaning the surface neatly till RCC Slabs without any mortar lumps,Mortar splashings etc complete before treatment.				
	b) Providing & Laying with maximum 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag of cement, Including laying in required position in flooring. Curing the surface for 7 days for water tightness.				
	c) Laying brick bats with mortar using broken bricks/brick bats 25 mm to 115mm size with 50% of cement mortar 1:4 admixed with water proofing compound conforming to IS : 2645 and approved by SBI/APMC , Followed by 20 mm thick layer of cement mortar of mix 1:4 admixed with water proofing compound conforming to IS : 2645 and approved by SBI/APMC to required slope and treating similarly the adjoining walls upto 300 mm height including rounding of junctions of walls and slabs.				
	d)Application of Primer Providing and applying single coat of epoxy resin based acrylic primer cum sealer Maxshield® Primer at an coverage of 5 - 6 sqm per ltr depending upon surface before further application of waterproofing coat or as per manufacturers' specification. Application of Waterproofing Coating Providing and applying two coats of single component solvent based polyurtehane waterproofing coating Maxshield® PF 900 reinforced with nonwoven geotextile fabric of 60 GSM Maxshield® FABRIC soaked in Maxshield® PF 900 at an coverage of 2 kg per sqm including curing as per manufacturers' specification.				
	e) After curing the surface by ponding for 5-7 days, applying a second coat of cement slurry using 2.75kg/Sq m of cement admixed with water proofing compound conforming to IS:2645 and approved by SBI/APMC.				
	f) Finishing the surface with 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag, giving proper slops, compaction, curing, finishing preparing bell mouth etc. approved by SBI/APMC. along with Kani plaster over Round portion.(i.e. Vatta)				
	g) The whole terrace so finished shall be flooded with water uptill height of rounded portion for a minimum period of two weeks for curing and for final test. All above operations to be done in order and as directed and specified by the SBI/APMC.				
	Waterproofing to Terraces	Sqm	4,010	3,000	1,20,30,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
b	Waterproofing to Terraces using Brick bat coba and China mosaic				
	Providing and laying in-position integral cement based water proofing treatment including preparation of surface as required for treatment of roofs, balconies, terraces, top terraces etc consisting of following operations:				
	a) Applying a slurry coat of neat cement using 2.75 kg/Sq m. of cement admixed with water proofing compound conforming to IS. 2645 and approved by SBI/APMC over the RCC slab including adjoining walls upto 300 mm height including cleaning the surface neatly till RCC Slabs without any mortar lumps,Mortar splashings etc complete before treatment.				
	b) Providing & Laying with maximum 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag of cement, Including laying in required position in flooring. Curing the surface for 7 days for water tightness.				
	c) Laying brick bats with mortar using broken bricks/brick bats 25 mm to 115mm size with 50% of cement mortar 1:4 admixed with water proofing compound conforming to IS : 2645 and approved by SBI/APMC , Followed by 20 mm thick layer of cement mortar of mix 1:4 admixed with water proofing compound conforming to IS : 2645 and approved by SBI/APMC to required slope and treating similarly the adjoining walls upto 300 mm height including rounding of junctions of walls and slabs.				
	d)Application of Primer Providing and applying single coat of epoxy resin based acrylic primer cum sealer Maxshield® Primer at an coverage of 5 - 6 sqm per ltr depending upon surface before further application of waterproofing coat or as per manufacturers' specification. Application of Waterproofing Coating Providing and applying two coats of single component solvent based polyurtehane waterproofing coating Maxshield® PF 900 reinforced with nonwoven geotextile fabric of 60 GSM Maxshield® FABRIC soaked in Maxshield® PF 900 at an coverage of 2 kg per sqm including curing as per manufacturers' specification.				
	e) After curing the surface by ponding for 5-7 days, applying a second coat of cement slurry using 2.75kg/Sq m of cement admixed with water proofing compound conforming to IS:2645 and approved by SBI/APMC.				
	f) Finishing the surface with 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag, Including laying in required position flooring of broken China Mosaic (broken pieces of China Chips) of approved colour and pattern including neat cement float, giving proper slops, compaction, curing, finishing preparing bell mouth etc. approved by SBI/APMC. along with Kani plaster over Round portion.(i.e. Vatta)				
	g) The whole terrace so finished shall be flooded with water uptill height of rounded portion for a minimum period of two weeks for curing and for final test. All above operations to be done in order and as directed and specified by the SBI/APMC.	Sqm	590	3,700	21,83,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
57	Roof slab (Protection screed)				
	Protective Plaster: providing and laying avg. 20 mm thick, CM 1:4 plaster admixed with Integral Waterproofing Compound WP 10 of equivalent @ 150 ml / bag of cement. (cement and sand shall be supplied by the main contractor)	Sqm	4,420	600	26,52,000
58	Waterproofing treatment to Podium Slab				
	Providing and laying in-position integral cement based water proofing treatment including preparation of surface as required for treatment of roofs, balconies, terraces, top terraces etc consisting of following operations:				
	a) Applying a slurry coat of neat cement using 2.75 kg/Sq m. of cement admixed with water proofing compound conforming to IS. 2645 and approved by SBI/APMC over the RCC slab including adjoining walls upto 300 mm height including cleaning the surface neatly till RCC Slabs without any mortar lumps,Mortar splashings etc complete before treatment.				
	b) Providing & Laying with maximum 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag of cement, Including laying in required position in flooring. Curing the surface for 7 days for water tightness.				
	c) Laying brick bats with mortar using broken bricks/brick bats 25 mm to 115mm size with 50% of cement mortar 1:4 admixed with water proofing compound conforming to IS : 2645 and approved by SBI/APMC , Followed by 20 mm thick layer of cement mortar of mix 1:4 admixed with water proofing compound conforming to IS : 2645 and approved by SBI/APMC to required slope and treating similarly the adjoining walls upto 300 mm height including rounding of junctions of walls and slabs.				
	d)Application of Primer Providing and applying single coat of epoxy resin based acrylic primer cum sealer Maxshield® Primer at an coverage of 5 - 6 sqm per ltr depending upon surface before further application of waterproofing coat or as per manufacturers' specification. Application of Waterproofing Coating Providing and applying two coats of single component solvent based polyurtehane waterproofing coating Maxshield® PF 900 reinforced with nonwoven geotextile fabric of 60 GSM Maxshield® FABRIC soaked in Maxshield® PF 900 at an coverage of 2 kg per sqm including curing as per manufacturers' specification.				
	e) After curing the surface by ponding for 5-7 days, applying a second coat of cement slurry using 2.75kg/Sq m of cement admixed with water proofing compound conforming to IS:2645 and approved by SBI/APMC.				
	f) Finishing the surface with 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag, giving proper slops, compaction, curing, finishing preparing bell mouth etc. approved by SBI/APMC. along with Kani plaster over Round portion.(i.e. Vatta)				
	g) The whole podium so finished shall be flooded with water uptill height of rounded portion for a minimum period of two weeks for curing and for final test. All above operations to be done in order and as directed and specified by the SBI/APMC.	Sqm	1,000	3,000	30,00,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
59	Waterproofing Treatment to Toilet, Utility area and other Internal Wet areas				
	Single Component Polymer based Elastomeric Seamless Membrane.				
	Cleaning the surface in such a way that RCC slab appears clearly & same shall be free from mortar splashing, Lumps etc., thereafter providing and applying a single pack waterborne eco-friendly elastomer copolymer waterproof coating membrane Mfd. by M/s. Pidilite Industries Ltd., BASF India Ltd., Sunanda speciality coatings, Cika or equivalent. to be applied by brush or roller in 2 coats to achieve average thickness of 600 -800 microns with a total consumption of approx. 0.80 Litre/Sq m, having a tensile strength of ≥ 2.5 N/mm ² , & same shall have a Solid Content (w/w) $\geq 65\%$. , having elongation of minimum 300%, with adhesion of minimum 0.5 N/mm ² , with 2.00mm crack bridging capacity, with resistance to water penetration, for wet area application, Same to be applied over the horizontal and vertical wall surface etc., complete.				
	Providing & Laying with maximum 20 mm thick cement mortar of mix 1:4 (1cement :4 coarse sand) admixed with water proofing compound (conforming to IS : 2645) at the rate of 1 Kg per bag of cement, Including laying in required position in flooring. Curing the surface for 7 days for water tightness. also includes sealing sleeves, outlets with non shrink, free flow, high strength cementitious grout material to seal the peripheral gap between pipe and waterproofing screed.				
	Protective Epoxy Bond Coat				
	Providing and applying a bond coat of Pidilite/ Sika/Basf/ Sunanda or equivalent, as a bond coat in one layer, followed by sprinkling of Dry natural sand while the bond coat is still tacky so as to create a rough Key for the subsequent layer etc. complete.				
		Sqm	2,360	950	22,42,000
60	LIFT, UG AND OHT Tanks				
	Providing and laying waterproofing and protective tank lining system through approved specialized agencies for U/G & O/H storage tank internally consisting of cleaning the surface of the internal concrete surface of the floor as well as walls thoroughly by wire brushing and washing with high pressure water jet, to remove any loose laitance as well as contaminates, thereafter inspecting the concrete surface to identify any honeycombs, crack, voids or defects in the concrete such as cracks, construction joints etc., and sealing all such locations with with solvent free epoxy mortar, using epoxy resin and hardener. Thereafter providing and applying containment membrane lining system consisting of combination of epoxy and polysulphide formulation. Epoxy Polysulphide formulation membrane lining cures to produce a tough, flexible coating system with excellent abrasion and chemical resistance and waterproof properties and consists of 2 component, which shall be applied in 2 coats by way of spray or roller to provide a minimum wet film thickness of 0.25 mm per coat with material consumption of approx. 0.3 Kg per Sq mt. per coat, all as per manufacturers recommendation etc. Complete. Treatment used should be non toxic system, to be used in drinking water and potable water tanks, tested to BS 6920.				
		Sqm	1,450	1,500	21,75,000
TOTAL OF SECTION - XII					2,42,82,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
XIII	MISCELLANEOUS				
61	CLADDING IN ELEVATION				
	Providing ,Supply and installation n of 6mm thick Exterior grade Quality panels of standard size of 1300 x 2800 mm.				
	Installation of these panels with Wall Bracket with Wind load and Dead load slot as per design, anchored by standard Hilti/Fischer or equivalent make anchor fasteners along with Thermal Separator.				
	Aluminium panels must be done by MBE Rivets (Fixed and Sliding Points) with minimum recommended Aluminum T section of 75x37x2 mm thick and L section 40x37x2.8mm thick.				
	The panels should be with both side décor and duromer high-pressure laminates (HPL) as per EN 438-6 type EDF with norm conformity as per EN 438- 7 manufactured using NT Technology (Non-fading high performance acrylic polyurethane surface technology) conforming to norms with CE Mark. Panels are double hardened including acrylic polyurethane resin which is thermally cured under high pressure. The panel should have passed salt spray test for 24 hours, possess performance sustainability certificate for more than 33 years and proof of experience in handling similar type of work in India for more than 10 years. Material should be class B / class 1 complying with NBC 2016, ASTM E 84, BS 476 - 7.				
	Installation of the HPL Panels will be done using Rear Ventilated Principles only which is ensured by providing gaps of 20 mm at top and bottom(for free flow of air behind the façade) for the façade and using the framework with no horizontal section. Approved make : Fundermax or approved equivalent	Sqm	1,880	12,000	2,25,60,000
62	VERTICAL ALUMINIUM LOUVER				
	Providing and fixing in position powder coated hollow aluminium section Glossy/Matt finish as per Architect approved colour and make with 1.5mm Thk Gauge Aluminium hollow section of size 75x40 mm as vertical members fixed on horizontal support sections of 50x100mm over the wall / structural members @ 450mm c/c. The spacing between vertical fins shall be minimum 100mm c/c. Rates shall be including cutting, bending as per profile, screwing, including all Necessary additional supports, Fittings & Fixtures, lead lift scaffolding etc completed as directed as per the desgin and colour given (Champagne Gold) by the SBI/APMC. (For payment only front /elevation area measurement considered)	Sqm	2,690	6,917	1,86,06,730
63	MANHOLE COVER				
	Providing and fixing in position Heavy duty CI Circular heavy duty, manhole cover of IS 1726 EHD 35 with frame. 600 mm dia of 208 kg/ no including fixing with CC 1:2:4 in position of approved colour Shreeji /Burhani/Plasma or Equivalent approved make.	Nos	30	23,100	6,93,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
64	MS RUNGS IN WATER TANKS				
	Providing & Fixing 20mm dia epoxy coated steel bars as Rungs of size 200 x 200 in Over Head Water Tank & Under Ground Water tank to be fitted at the time of casting RCC walls for Tanks including cost for inserting rungs on shuttering through holes and properly tying with the reinforcement bars of the tank (as per structural drawings/ details) with required spacing of maximum 300m between rungs or lesser as per site conditions.				
		Nos	130	500	65,000
65	EXPANSION WALL JOINT				
	Providing and fixing of expansion joint system related with wall joint (internal/external) location as per drawings and direction of Engineer-In- Charge. The joints shall be of extruded aluminum base members, self aligning / centering arrangement and support plates as per ASTM B221-02. The material shall be such that it provides an Expansion Joints System suitable for vertical wall to wall/ wall to corner application, both new and existing construction in office Buildings & complexes with no slipping down tendency amongst the components of the Joint System. The Joint System shall utilize light weight aluminum profiles exhibiting minimal exposed aluminum surfaces mechanically snap locking the multi- cellular to facilitate movement. (Material shall confirm to ASTM 6063.)				
	Wall Joint of 100 mm gap	Rm	370	6,300	23,31,000
66	EXPANSION FLOOR JOINT				
	Providing and fixing of expansion joint system related with floor location as per drawings and direction of Engineer-In-Charge. The joints system will be of extruded aluminum base members, self aligning / self centering arrangement and support plates etc. as per ASTM B221-02. The system shall be such that it provides floor to floor /floor to wall expansion control system for various vertical location in load application areas that accommodates multi directional seismic movement without stress to it's components. System shall consist of metal profiles with a universal aluminum base member designed to accommodate various project conditions and finish floor treatments.				
	The cover plate shall be designed of width and thickness required to satisfy projects movement and loading requirements and secured to base members by utilizing manufacturer's pre-engineered self-centering arrangement that freely rotates / moves in all directions. The Self – centering arrangement shall exhibit circular sphere ends that lock and slide inside the corresponding aluminum extrusion cavity to allow freedom of movement and flexure in all directions including vertical displacement. Provision of Moisture Barrier Membrane in the Joint System to have watertight joint is mandatory requirement all as per the manufactures design and as approved by Engineer -in-Charge . (Material shall confirm to ASTM 6063.)				
	Floor Joint of 100 mm gap	Sqm	140	700	98,000

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
67	PAVERS- 80 MM THICK FACTORY MADE CEMENT CONCRETE INTERLOCKING PAVER BLOCK				
	Providing and laying 80 mm thick factory made cement concrete interlocking paver block of M -30 grade made by block making machine with strong vibratory compaction, of approved size, design & shape, laid in required colour and pattern over and including 50mm thick compacted bed of coarse sand, filling the joints with fine sand etc. all complete as per the direction of SBI/APMC. Make - Super, Nitco, Bharat	Sqm	2,060	1,000	20,60,000
68	RCC SUPER STRUCTURE DISMANTLING				
	Dismantling manually or by mechanical means all RCC members in super structure viz columns, beams, slab, Pardis, RCC walls, water tanks etc. with all related equipment, at all height and location. The steel in the RCC structures shall be separated from the concrete and stack-up separately. The contractor should take care that dismantling of structure should not cause any damage to the existing structures near the building and even take care of dust control, noise control etc. Blasting is not permitted. All the work shall be carried out as instructed by Architect / SBI. The rate includes cutting of MS reinforcement steel bars using mechanical cutter and stacking the same as directed at site and carting away unserviceable material / debris to a place approved by Municipal Corporation.	Cum	30	4,100	1,23,000
69	DISMANTLING OF STONE / BLOCK / BRICK MASONRY				
	Dismantling manually or by mechanical means all type of Stone / brick / block masonry in the entire building including plastered walls, parapet on terrace, compound wall etc. in existing sub structure and superstructure at all height and location. The contractor should take care that dismantling of structure should not cause any damage to the existing / adjacent structures near the building and even take care of dust control, noise control etc. Blasting is not permitted. All the work shall be carried out as instructed by Architect / SBI. The rate to include carting away unserviceable material / debris to a place approved by Municipal Corporation. The work also includes levelling the ground after carting away.	Cum	20	2,250	45,000
70	PERCULATION PIT				
	Providing and laying percolation pit of 100mm dia at 6 meter at Center to center below the storm water drain to depth of 3 meter (Top level of PVC pipe should be flush with the drain level) , with perforated slotted PVC pipe 100 mm dia filled with 4 layers of pebbles, fine aggregate, coarse aggregate of sizes 40mm, 20mm, 10mm and 7.5mm equally. All the work shall be carried out as instructed by Architect / SBI. The bottom of the PVC pipe inserted shall be closed using cap of required size.	Nos	54	12,900	6,96,600

Sect/ Item No.	Description	Unit	Total Quantity	Rate	Amount
71	READY MADE GUARD CABIN				
	Providing and fixing Readymade PVC Guard closed cabin of size - 10' x 8' x 7' with counter top and ledge out side . Cabin insulated with Puff Panel including attached toilet and Phe works as required.	Nos	1	2,50,000	2,50,000
72	BORE WELL				
	Drilling of 10" dia vertical borewell to a depth of 12 - 15 m so as to have a finished size of 6" dia borewell in boulder area to any required depth , including Transportation from one place to another place by direct or reverse rotary mud circulation method using Contractor's rig tools, fuel, labourers, bentonite mud and water required for drilling at the site shown				
	Supply and laying following 6" dia PVC casing pipe with excise duty as per IS 12818 / 1992				
	Labour charges for inserting PVC casing pipes assembly (with slots or without slots) in the drilled hole including jointing the pipes with PVC couplers with cement paste etc., complete, supply and packing the annular space with pebbles of size and quality as approved . Supplying and filling with river pebbles in annular space between bore and PVC pipe up to hard rock. Charges for developing the borewell with air compressor of 300 CFM capacity (Minimum 8 Hours) including transportation labour and fuel charges for compressor				
	Charges for developing the borewell with air compressor of 300 CFM capacity (Minimum 8 Hours) including transportation labour and fuel charges for compressor				
	Supply ,erection and testing & commissioning of 1.5HP submersible pumpset in the to suit 150mm borewell . total head of 180 Ft pumps fitted with bronze impellers and having delivery size of 40mm - 430V along with 3mts cable and manufactures standard control panel which is mandatory and as directed by the officer incharge and other required supporting items like SS hose nipple, nylon rope, clamps, bolt and nuts, GI nipple, 50mm dia HDPE hose of required length etc completed and as directed by the officer incharge.	L.S	2	2,50,000	5,00,000
	TOTAL OF SECTION - XIII				4,80,28,330
	TOTAL AMOUNT				92,28,46,030

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI.					
S.No	Description	UNIT	TOTAL QTY	RATE	AMOUNT
	PART F: BOQ FOR PLUMBING & SANITARY WORKS				
A	SANITARY INSTALLATIONS				
	Approved Makes of Sanitary Fixtures & CP Fittings - Jaquar/Hindware/Parryware.				
1	European wall mounted with concealed cistern				
	Supplying, fixing, testing and commissioning of approved make white colour European wall mounted with concealed cistern The quoted price shall also include installation of CI brackets/ chair brackets as per site conditions for fixing closet , 1no. CP angle cock, CP wall flanges, plastic solid seat cover with rubber buffers and flap, Heavy duty plastic hinges, CI bracket / Pedestal etc., complete. Basic cost of closet =8000 & concealed cistern with flush plate = 4500,Closet : Parryware make Cascade NXT C0207 Concealed cistern : Parryware make, Flush plate: Parryware make e8221a1 or equivalent approved makes (for each fittings)	Nos	311.00	26,000.00	80,86,000.00
2	Washbasin above counter with Pillar cock				
	Supplying, installing, testing and commissioning approved make wash basin white colour fixed above counter with 32mm dia CP waste coupling, 32mm dia C.P.bottle trap with C.P. extension pipe, 1 No.15mm dia extended C.P. Pillar cock, 1 No.15mm dia C.P. angle cock, 1 No. C.P. flexible inlet connection pipe G.I. expansion bolts etc., complete including silicon sealant. Basic cost of wash basin = 5000 & Pillar Cock = 2500, Wash basin : Parryware make Cascade NXT C0402, CP Fittings : Jaquar Make Waste coupling – ALD 709, Bottle Trap – 769, Angle cock – 059KN, Jaquar Make FUS-29005NB or equivalent approved makes (for each fittings)	Nos	311.00	14,000.00	43,54,000.00
3	Washbasin below counter with Pillar cock				
	Supplying, installing, testing and commissioning approved make wash basin white colour fixed below counter with 32mm dia CP waste coupling, 32mm dia C.P.bottle trap with C.P. extension pipe, 1 No.15mm dia C.P. Pillar cock, 1 No.15mm dia C.P. angle cock, 1 No. C.P. flexible inlet connection pipe G.I. expansion bolts etc., complete including silicon sealant. Basic cost of wash basin = 3500 & Pillar Cock = 2100,. Wash basin : Parryware make Cascade NXT C0407 CP Fittings : Jaquar Make Waste coupling – ALD 709, Bottle Trap – 769, Angle cock – 059KN, Jaquar Make FUS-29023B or equivalent approved makes (for each fittings)	Nos	1.00	13,800.00	13,800.00
4	Urinal with Pushcock				
	Supplying,fixing, testing and commissioning approved make white colour large flat back urinals with 40mm C.P. dome type waste coupling 40mm C.P. bottle trap with C.P. extension pipe, 15 mm dia Pressmatic push cock with auto closing or angle cock etc. valve, C.P. spreader unit GI expansion bolts etc., complete , Basic cost of Urinal = 5000 & Pushcock = 1500, Urinal : Parryware make New Magnum, CP Fittings : Jaquar Make, push cock – PRS-CHR-077, Waste coupling – ALD 709, Bottle Trap – 769, or equivalent approved makes (for each fittings)	Nos	4.00	25,000.00	1,00,000.00
5	SS Sink with Drainboard				
	Supplying, Installing, testing and commissioning of approved make stainless steel kitchen sink with drain board of Overall size 950 x510mm, Bowl Size of 410x410x230mm, CP waste coupling, 40mm dia flexible pipe, 1 no. 15mm dia sink cock table/wall mounted type with swinging spout, C.I.brackets etc.,complete. Single bowl - Nirali / Franke make Basic cost of Sink = 5000 & Pillar Cock = 2500, CP Fittings : Jaquar Make FLR 5347N or equivalent approved makes (for each fittings)	Nos	128.00	15,950.00	20,41,600.00
6	Shower with Single lever divertor				
	Supplying, Installing, testing and commissioning approved make 15mm CP shower unit comprising of CP 15mm dia single lever divertor with CP spout, overhead shower set comprising of CP shower arm, shower rose 150mm dia etc.,complete with CP wall flanges. Basic cost of set = 10500, CP Fittings : Jaquar Make Single lever Divertor - FLR-5079K & ALD-079 Shower Rose OHS -497N / OHS-1805, Shower Arm SHA-483 / SHA-479 L 300 , Spout – SPJ-5429 or equivalent approved makes (for each fittings)	Nos	310.00	18,150.00	56,26,500.00
7	Health Faucet				
	Supplying & fixing of approved make 15mm CP health faucet with one no. of CP angle cock, CP flexible tube 1.0m long, CP crutch, nozzle etc., complete. Basic cost = 2050 CP Fittings : Jaquar Make ALD-565 or equivalent approved makes	Nos	311.00	2,500.00	7,77,500.00
8	Two way Bib cock				
	Supplying & fixing of approved make 15mm CP two way bib cock with necessary CP wall flanges etc., complete. Basic cost = 1800. CP Fittings : Jaquar Make FUS-29041 or equivalent approved makes	Nos	311.00	2,500.00	7,77,500.00
9	Towel Ring				
	Supplying and fixing approved make C.P. towel ring with necessary fixing materials. Basic cost = 850, CP Fittings : Jaquar Make AQN-7721 or equivalent approved makes	Nos	1.00	1,320.00	1,320.00
10	Towel Rail				
	Supplying and fixing approved make C.P. Towel Rail with necessary fixing materials etc. Basic cost = 1200 CP Fittings : Jaquar Make AQN-7711 or equivalent approved makes	Nos	311.00	1,820.00	5,66,020.00
11	Soap Dish				
	Supplying & fixing of approved make soap dish with necessary fittings. Basic cost = 650, Jaquar Make ACN-1131N or equivalent approved makes	Nos	310.00	1,160.00	3,59,600.00
12	Toilet Paper Holder				
	Supplying and fixing approved make stainless steel paper holder with necessary fixing material paper roll etc.complete. Basic cost = 850 Jaquar Make ACN-1151N or equivalent approved makes	Nos	311.00	1,150.00	3,57,650.00

13	Washing Machine Bib Cock Supplying & fixing of approved make washing machine bib cock with necessary fittings. Basic cost = 850, Jaquar Make CON-049NKN or equivalent approved makes	Nos	128.00	1,320.00	1,68,960.00
14	Concealed Stop Cock Supplying & fixing of approved make concealed stop cock inside toilet with necessary fittings. Basic cost = 1850 Jaquar Make or equivalent approved makes	Nos	615.00	2,320.00	14,26,800.00
15	Angle Cock Supplying & fixing of approved make angle cock for aqua guard, geyser provision and washing machine, Basic cost = 550, Jaquar Make FUS-29053 or equivalent approved makes	Nos	870.00	850.00	7,39,500.00
16	Mirror Supplying and fixing 800x600mm beveled edge mirror of superior glass (of approved quality) complete with 6 mm thick Ply board ground fixed to wooden cleats with C.P. brass screws and washers complete.	Nos	312.00	2,500.00	7,80,000.00
TOTAL OF SECTION -A					2,61,76,750.00
B INTERNAL SOIL & WASTE SYSTEM:					
UPVC CLASS III PIPES					
Supplying, laying, jointing, testing and commissioning of approved make UPVC pipes class III 6 Kg/cm2 with solvent cement joints conforming to IS 4985 for Wash basin, Washing machine & Sink outlets. The quoted rate shall include necessary chisling in walls making good the same in C.M. 1:4, encasing the pipes in toilet sunken /below false ceiling and under ground in M-15 as per specifications 100 mm thk. all around with necessary shuttering arrangement, necessary fittings viz., elbows, tees etc. complete.					
17	40 mm dia	Rmt	685.00	270.00	1,84,950.00
18	50 mm dia	Rmt	290.00	350.00	1,01,500.00
19	63 mm dia	Rmt	75.00	420.00	31,500.00
PVC SWR GRADE PIPES					
Supplying, laying, jointing, testing and commissioning of approved make PVC pipes SWR grade conforming to IS 13592 Class "B" with rubber ring joints conforming to IS 5382 for exposed piping in shaft, solvent jointing for Suspended piping in ceiling Lvl. The quoted rate shall include necessary fittings like Tees, 45 or 90 deg. Bends, Couplers, Offsets, etc., G.I. brackets with pvc coated GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc., complete.					
20	75 mm dia (6 Kg/cm2)	Rmt	1,900.00	745.00	14,15,500.00
21	110 mm dia (6 Kg/cm2)	Rmt	925.00	1,200.00	11,10,000.00
PVC Floor drain					
22	Supplying, laying, jointing, testing and comissioning of approved make PVC floor drain with C P grating with required concrete encasing all around, complete.	Nos	415.00	630.00	2,61,450.00
PVC Floor Trap					
23	Supplying, fixing and testing of approved make PVC Multi floor trap (heavy duty hinged type) with brass CP frame & cover (150 mm x 150 mm square / 125 mm round). Trap of 100 mm dia. Inlet and 75 mm dia outlet fixed on a bed of cement concrete in sunken area or necessary PVC pipe sleeve for floor trap inside toilet slab & required concrete encasing all around, complete.	Nos	904.00	1,250.00	11,30,000.00
Core Cutting					
Drilling with core cutting machine in RCC beams, slabs, floors (200mm deep) etc., for laying pipes and rendering the same in RCC 1:2:4, finishing the same to the satisfaction of the owner or his authorised representative etc., including nominal reinforcement wherever required complete.					
24	110 mm dia	Nos	25.00	650.00	16,250.00
25	125 mm dia	Nos	1,265.00	800.00	10,12,000.00
TOTAL OF SECTION - B					52,63,150.00
C VERTICAL SOIL, WASTE & RAIN WATER PIPE SYSTEM:					
PVC SWR GRADE PIPES					
Supplying, laying, jointing, testing and commissioning of approved make PVC pipes SWR grade conforming to IS 13592 Class "B" with rubber ring joints conforming to IS 5382 for exposed piping in shaft .The quoted rate shall include necessary fittings like Tees, 45 or 90 deg. Bends, Couplers, Offsets, etc., G.I. brackets with pvc coated GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc., complete.(External piping in shaft & ceiling lvl). , as directed by Engineer-in-Charge.					
26	75 mm dia (6kg/cm2)	Rmt	1,470.00	745.00	10,95,150.00
27	110 mm dia (6kg/cm2)	Rmt	1,042.00	1,200.00	12,50,400.00
28	160 mm dia (6kg/cm2)	Rmt	350.00	1,550.00	5,42,500.00
29	200 mm dia (6kg/cm2)	Rmt	84.00	1,950.00	1,63,800.00

	PVC SWR GRADE PIPES				
	Supplying, jointing, testing & Commissioning of approved make PVC pipe SWR grade B confirming to IS13592 with rubber ring joints for rain water disposal confirming to IS 5382. The quoted rate shall include necessary fittings like Tees, 45 or 90 deg. Bends, , Offsets etc., G.I. brackets with pvc coated GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc., complete. (External piping in shaft & ceiling lvl)				
30	75 mm dia (Rain Water pipe 4 Kg/cm2)	Rmt	1,440.00	715.00	10,29,600.00
31	110 mm dia(Rain Water pipe 4 Kg/cm2)	Rmt	257.00	1,110.00	2,85,270.00
32	160 mm dia(Rain Water pipe 4 Kg/cm2)	Rmt	380.00	1,385.00	5,26,300.00
33	200 mm dia(Rain Water pipe 4 Kg/cm2)	Rmt	176.00	1,830.00	3,22,080.00
	RAIN WATER PVC PIPE FILTER				
34	Supplying and fixing of Jal santosh/Supreme/Ashirward/equivalent PVC pipe filter for rain water harvesting with necessary inlet & outlet valves, stainless steel strainer filtering rainwater before entering into sump or borewell. The quoted rate shall include necessary fittings like Bends, , Offsets etc., G.I. brackets with pvc coated GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc., complete. as directed by Architect/ Bank.	Nos	30.00	12,500.00	3,75,000.00
	MS SUPPORTS				
35	Supplying and fixing MS slotted angle iron brackets,hangers, supports with necessary drilling holes to suit the insertion of pipe clamps deburning etc., as per detailed drawing for supporting soil, waste and rain water pipes and water supply pipes in shafts with suitable MS clamps, bolts, anchor bolts, nuts and washers fixed on shaft walls / outer walls etc. The quoted rate shall include with the cost of brackets galvanizing with zinc alloy of 0.08mm thick conformed to IS 1239 etc., complete as directed by the Architect/ PMC.	KG	100.00	200.00	20,000.00
	MS GRIP BOLTS / CLAMPS				
36	Supplying and fixing MS grip bolts / clamps in columns / beams / walls ceiling for fixing soil, waste and rain water pipe supports.	Nos	100.00	100.00	10,000.00
	TOTAL OF SECTION - C				56,20,100.00
	D EXTERNAL SANITARY SYSTEM:				
	PVC CLASS III PIPES				
	Supplying, laying, jointing, testing and commissioning approved make PVC pipes class III 6 Kg/cm2 with solvent cement joints confirming to IS 4985 for Under ground drainage works. The quoted rate shall include necessary earth work, fittings, pipe to be laid below ground level in trenches upto required depth, dewatering, jointing of pipes, water testing and setting/encasing the pipes in cement concrete M-15 as per specifications 150mm thk. all around, refilling, and making good the same complete as required.				
37	110 mm dia (6kg/cm2)	Rmt	10.00	1,200.00	12,000.00
38	160 mm dia (6kg/cm2)	Rmt	80.00	1,550.00	1,24,000.00
	GULLY TRAP				
39	Supplying, fixing, testing and commissioning of glazed stoneware gully traps, 300x300mm, 100 mm dia outlet set in PCC 1:4:8 bed, 115 thick masonry chamber in CM 1:4, internal and external plastering 12mm thick in CM 1:6, necessary earthwork, C.I. cover (medium duty) etc., complete.	Nos	4.00	5,720.00	22,880.00
	INSPECTION CHAMBER				
	Providing and constructing inspection chambers comprising burnt brick/solid block masonry walls of 230 mm thk., built in CM 1:4 over M-15 CC bed of thickness 100 mm ,Heavy duty SFRC Covers, benching in CC M-15, internal faces plastered smooth in CM 1:4, 15mm thick and outside Rough Plastered with CM 1:6, 15mm thick including necessary earth work excavation returning and refilling and disposing surplus soil as directed by the Engineer in charge.				
40	600x600 mm size internal and depth more than 450 mm but not exceeding 1000 mm	Nos	4.00	19,800.00	79,200.00
41	900x900 mm size internal and depth more than 1200 mm but not exceeding 1500 mm	Nos	2.00	22,550.00	45,100.00
	EXTRA DEPTH				
42	Extra depth more than 1000 mm for 600 x600mm chamber	Rmt	4.00	6,900.00	27,600.00
43	Extra depth more than 1500 mm for 900 x 900mm chamber	Rmt	2.00	8,900.00	17,800.00
	EXCAVATION				
44	Excavation in trenches in ordinary hard soil (other than in rock requiring chiselling and blasting) for laying Sewer lines and refilling of trenches after pipes are laid in layers not exceeding 20 cm, watering and consolidation and disposing off surplus earth as directed by the Engineer incharge, providing necessary Shoring & Strutting while excavating etc., complete Depth of excavation up to 1.2m	Cum	30.00	440.00	13,200.00
	TOTAL OF SECTION - D				3,41,780.00
	E EXTERNAL RAIN WATER SYSTEM:				
	RAIN WATER CATCH BASIN				
	Construction of catch basin of size mentioned below upto 600mm clear internal dimension for storm water with 230mm thick walls in well burnt table moulded bricks in CM 1:4 over a bed of 100 mm thick PCC 1:4:8 (M10) internal walls plastered smooth in CM 1:3 and external walls plastered in 1:3 with sponge finish. Fixing precast Heavy Duty RCC grating of size 400 x400mm on top surface flushed to the finished floor level with necessary excavation, back filling the selected excavated earth with all leads and lifts etc., complete				
45	600x600 mm size internal and depth more than 450 mm but not exceeding 1000 mm	Nos	20.00	21,000.00	4,20,000.00
46	900x900 mm size internal and depth more than 1200 mm but not exceeding 1500 mm	Nos	5.00	25,000.00	1,25,000.00

	RCC HUME PIPE				
	Supplying, installing, testing and commissioning of R.C.C NP2 class Hume pipe laid to the required level and grade with collars joining in CM 1:3 and filling the excavated trench with approved quality saturated sand on sides, bottom and top surface of pipe (sand cushion at joints only) with all leads and lifts, making necessary connections as required etc., and Including Excavation.				
47	200 mm dia	Rmt	250.00	1,080.00	2,70,000.00
48	250 mm dia	Rmt	100.00	1,270.00	1,27,000.00
	RAIN WATER RECHARGE PIT				
49	Providing & construction of Rain water recharge pit as per IS 24701 P1& II and relevant construction drawing. The pit shall be construction of RCC Rings with RCC top slab and PCC bed at bottom and Recharge pit filling of 40mm jelly & sand Size of Recharge pit:- 1.2m dia x 2.4mdepth	Nos	10.00	24,200.00	2,42,000.00
	EXCAVATION				
50	Excavation in trenches in ordinary hard soil (other than in rock requiring chiselling and blasting) for laying Sewer lines and refilling of trenches after pipes are laid in layers not exceeding 20 cm, watering and consolidation and disposing off surplus earth as directed by the Engineer incharge, providing necessary Shoring & Strutting while excavating etc., complete	Cum	30.00	440.00	13,200.00
	TOTAL OF SECTION - E				11,97,200.00
	F INTERNAL & VERTICAL (SHAFT) WATER SUPPLY SYSTEM:				
	CPVC SDR 11 PIPES				
	Supplying, installing, fixing, jointing, testing and commissioning of approved make Chlorinated Polyvinyl Chloride pipes (CPVC) SDR.11, ASTM CTS, ASTM IPS. The fittings and specials such as tees, elbows, couplers, bends, enlargers etc., with CPVC brass threaded combination / transition specials such as male adapters brass threaded female adapters, brass FPT Tee, Brass FPT elbow etc., where connection with metal is to be made including necessary drilling holes, chasing walls and making the same good in geru mixed cement mortar 1:1 restore the same original condition neatly as directed by the Engineer incharge. The quoted rate shall include G.I. brackets with GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc. and pressure testing (Once at the time of fixing the concealed pipes and once pre tiling). Joints to be made with CPVC solvent cement up to 50mm dia as per ASTM D2846 and above 50mm dia primer P70, and solvent confirming to ASTM F4931 / 724.(Cold, Hot & RO water supply)				
51	20 mm dia	Rmt	5,835.00	400.00	23,34,000.00
52	25 mm dia	Rmt	540.00	645.00	3,48,300.00
53	32 mm dia	Rmt	490.00	700.00	3,43,000.00
54	40 mm dia	Rmt	586.00	930.00	5,44,980.00
55	50 mm dia	Rmt	946.00	990.00	9,36,540.00
56	63 mm dia	Rmt	258.00	1,100.00	2,83,800.00
	UPVC SCH 40 PIPES & SCH 80 FITTINGS				
	Supplying, installing, fixing, jointing, testing and commissioning of approved make sch 40 UPVC pipes and sch 80 fittings, ASTM CTS, ASTM IPS. The fittings and specials such as tees, elbows, couplers, bends, enlargers etc. with necessary fittings & suitable solvent/rubber ring joints. The quoted rate shall include G.I. brackets with GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc. Complete.				
57	50 mm dia (inlet to Flushing OHT)	Rmt	148.00	1,050.00	1,55,400.00
58	63 mm dia (inlet to Fire/Domestic OHT)	Rmt	347.00	1,400.00	4,85,800.00
	BALL VALVE				
	Supplying, installing, testing and commissioning of approved make full way lever operated forged brass ball valve of brass body with forged brass hard chromeplated steel ball tested to a pressure not less than 20 Kg / sqcm with threaded joints complete.				
59	20 mm dia	Nos	754.00	1,020.00	7,69,080.00
60	25 mm dia	Nos	68.00	1,240.00	84,320.00
61	50 mm dia	Nos	20.00	3,680.00	73,600.00
	BUTTERFLY VALVE				
	Supplying, installing, testing and commissioning of approved make butterfly valve (upto 150mm dia with hand lever operation & above with gear box operation). Tested to a pressure not less than 16 Kg/Sq.cm. Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers & painting complete as required				
62	65 mm dia	Nos	10.00	4,950.00	49,500.00
	PRESSURE REDUCING VALVE				
	Supplying, installing, testing and commissioning of approved make Gun Metal / Brass body Pressure Reducing Valve suitable for a maximum inlet pressure of upto 25 bars reduce to anywhere between 1 to 1.5 Bars, the valve incorporate inherent design advantage like reversible double reinforced food grade EPDM rubber diaphragm, pressure gauge connections on both sides of valves etc complete (Female Screwed Ends).				
63	20 mm dia	Nos	10.00	6,050.00	60,500.00
	TOTAL OF SECTION - F				64,68,820.00

G	TERRACE WATER SUPPLY SYSTEM				
	UPVC SCH 40 Pipes & SCH 80 Fittings				
	Supplying, installing, fixing, jointing, testing and commissioning of approved make sch 40 UPVC pipes and sch 80 fittings, ASTM CTS, ASTM IPS. The fittings and specials such as tees, elbows, couplers, bends, enlargers etc. with necessary fittings & suitable solvent/rubber ring joints. The quoted rate shall include G.I. brackets with GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc. Complete.				
64	25 mm dia	Rmt	45.00	590.00	26,550.00
65	32 mm dia	Rmt	35.00	660.00	23,100.00
66	40 mm dia	Rmt	55.00	750.00	41,250.00
67	50 mm dia	Rmt	75.00	1,050.00	78,750.00
68	63 mm dia	Rmt	125.00	1,400.00	1,75,000.00
69	80 mm dia (Domestic & Flushing terminal main on terrace)	Rmt	230.00	1,520.00	3,49,600.00
70	100 mm dia	Rmt	20.00	1,700.00	34,000.00
	BALL VALVE				
	Supplying, installing, testing and commissioning of approved make full way lever operated forged brass ball valve of brass body with forged brass hard chromeplated steel ball tested to a pressure not less than 20 Kg / sqcm with threaded joints complete.				
71	25 mm dia	Nos	15.00	1,240.00	18,600.00
72	32 mm dia	Nos	11.00	1,660.00	18,260.00
73	40 mm dia	Nos	45.00	1,900.00	85,500.00
	BUTTERFLY VALVE				
	Supplying, installing, testing and commissioning of approved make butterfly valve (upto 150mm dia with hand lever operation & above with gear box operation). Tested to a pressure not less than 16 Kg/Sq.cm. Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers & painting complete as required				
74	50 mm dia	Nos	54.00	3,580.00	1,93,320.00
75	65 mm dia	Nos	26.00	4,950.00	1,28,700.00
76	80 mm dia	Nos	24.00	5,570.00	1,33,680.00
77	100 mm dia	Nos	8.00	7,210.00	57,680.00
	NON RETURN VALVE				
	Supplying and fixing Non return valve of approved make with ISI make including fixing in position, making holes, restoring the damaged portions to original position and applying jointing compound at threaded ends and testing etc. complete as directed				
78	50 mm dia	Nos	4.00	10,060.00	40,240.00
79	65 mm dia	Nos	4.00	11,250.00	45,000.00
	SOLENOID VALVE				
	Supplying, installing, testing and commissioning approved make electrically operated solenoid valve with level controllers to be fixed in the OHT. The cost of valve shall also include necessary level switches, wiring upto control panel at terrace level, control panel, etc.,complete				
80	50 mm dia	Nos	4.00	8,360.00	33,440.00
81	65 mm dia	Nos	1.00	9,630.00	9,630.00
	PRESSURE REDUCING VALVE				
	Supplying, installing, testing and commissioning of approved make Gun Metal / Brass body Pressure Reducing Valve suitable for a maximum inlet pressure of upto 25 bars reduce to anywhere between 1 to 1.5 Bars, the valve incorporate inherent design advantage like reversible double reinforced food grade EPDM rubber diaphragm, pressure gauge connections on both sides of valves etc complete (Female Screwed Ends).				
82	32mm dia	Nos	6.00	8,500.00	51,000.00
83	40mm dia	Nos	42.00	9,350.00	3,92,700.00
84	50mm dia	Nos	50.00	12,650.00	6,32,500.00
85	65mm dia	Nos	14.00	14,270.00	1,99,780.00
	AIR RELEASE VALVE				
	Supplying and fixing brass air release valve of approved make with ISI make including fixing in position, making holes, restoring the damaged portions to original position and applying jointing compound at threaded ends and testing etc. complete as directed.				
86	25mm dia	Nos	15.00	2,420.00	36,300.00
	MASONRY PEDESTAL				
	Providing and constructing of masonry pedestals of size as mentioned below with bricks of class B50 designation in CM1:3 and finishing neat coat of floating cement and making good with necessary clamps, bolts and nuts etc.				
87	230x230x300mm	Nos	75.00	600.00	45,000.00
	TOTAL OF SECTION - G				28,49,580.00

H	EXTERNAL WATER SUPPLY SYSTEM				
	UPVC SCH 80 PIPES				
	Supplying, laying, jointing, testing & Commissioning of approved make uPVC Schedule 80 pressure pipes as per ASTM D 1785 with Sch 80 Fittings as per specifications, jointing with hold tite / super tite & teflon tape and complete with fittings such as sockets, unions, reducers long or short bends, elbows, nipples flanges, checknuts, etc. Pressure testing of piping and back filling of trench complete as required. The quoted rate shall include necessary earth works, fittings like Tees, Bends, Offsets etc., G.I. brackets with pvc coated GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc., complete for external pipes. (For under ground or above ground external water supply piping.)				
88	80 mm dia (Borewell water inlet to ug sump)	Rmt	50.00	1,800.00	90,000.00
89	63 mm dia	Rmt	20.00	1,370.00	27,400.00
90	50 mm dia	Rmt	20.00	1,145.00	22,900.00
91	25 mm dia	Rmt	50.00	730.00	36,500.00
	WATER METER				
	Supplying, installing, testing and commissioning approved make cast brass body water meter with necessary unions for fixing on the inlet mains as shown on the drawing etc, complete with necessary ball valves, unions, etc complete				
92	25 mm dia	Nos	2.00	10,580.00	21,160.00
	VALVE CHAMBER				
	Providing and constructing Valve Chamber comprising 230MM Thick burnt brick masonry walls built in CM 1:4 over CC bed ,DI cover slab benching in CC 1:2:4 internal faces plastered smooth in CM 1:4, 15mm thick and outside Rough Plastered with CM 1:6 15mm thick including 900x900mm medium earth work excavation returning and refilling and disposing surplus soil as directed by the Engineer in charge.				
93	600 x 600 x 600 mm depth	Nos	2.00	9,960.00	19,920.00
	EXCAVATION				
94	Excavation in trenches in ordinary hard soil (other than in rock requiring chiselling and blasting) for laying Sewer lines and refilling of trenches after pipes are laid in layers not exceeding 20 cm, watering and consolidation and disposing off surplus earth as directed by the Engineer incharge, providing necessary Shoring & Strutting while excavating etc., complete				
		Cum	30.00	440.00	13,200.00
	TOTAL OF SECTION - H				2,31,080.00
I	LANDSCAPE WATER SUPPLY SYSTEM				
	UPVC SCH 40 PIPES Sch 40 FITTINGS				
	Supplying, laying, jointing, testing & Commissioning of approved make uPVC Schedule 40 pressure pipes as per ASTM D 1785 with Sch 40 Fittings as per specifications, jointing with hold tite / super tite & teflon tape and complete with fittings such as sockets, unions, reducers long or short bends, elbows, nipples flanges, checknuts, etc. Pressure testing of piping and back filling of trench complete as required. The quoted rate shall include necessary fittings like Tees, Bends, Offsets etc., G.I. brackets with pvc coated GI clamps, nuts & bolts, anchor fasteners, grip bolts, GI threaded, rods, etc., complete for external pipes				
95	80 mm dia	Rmt	46.00	1,520.00	69,920.00
96	50 mm dia	Rmt	50.00	1,050.00	52,500.00
97	20 mm dia	Rmt	100.00	500.00	50,000.00
	BALL VALVE				
	Supplying, installing, testing and commissioning of approved make full way lever operated forged brass ball valve of brass body with forged brass hard chromeplated steel ball tested to a pressure not less than 20 Kg / sqcm with threaded joints complete.				
98	20 mm dia	Nos	15.00	1,020.00	15,300.00
	BUTTERFLY VALVE				
	Supplying, installing, testing and commissioning of approved make butterfly valve (upto 150mm dia with hand lever operation & above with gear box operation). Tested to a pressure not less than 16 Kg/Sq.cm. Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers & painting complete as required.				
99	80 mm dia	Nos	4.00	5,570.00	22,280.00
100	50 mm dia	Nos	4.00	3,580.00	14,320.00
	TOTAL OF SECTION - I				2,24,320.00

J	PUMPROOOM				
	GI PIPES				
	Supplying, laying, jointing, testing & commissioning of approved make class 'B' G.I. pipes and fittings confirming to IS 1239 PI & II, malleable fittings viz., tees, elbows, flanges, unions etc., including cutting threading, fixing terrace floor etc., as per drawings. The quoted rate shall include necessary making bores in walls, etc., a coat of including Painting.				
101	150 mm dia	Rmt	20.00	2,580.00	51,600.00
102	100 mm dia	Rmt	30.00	2,210.00	66,300.00
103	80 mm dia	Rmt	40.00	1,675.00	67,000.00
104	65 mm dia	Rmt	25.00	1,380.00	34,500.00
105	50 mm dia	Rmt	40.00	1,025.00	41,000.00
	BUTTERFLY VALVE				
	Supplying, installing, testing and commissioning of approved make butterfly valve (upto 150mm dia with hand lever operation & above with gear box operation). Tested to a pressure not less than 16 Kg/Sq.cm. Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers & painting complete as required				
106	100 mm dia	Nos	8.00	7,210.00	57,680.00
107	80 mm dia	Nos	8.00	5,570.00	44,560.00
108	65 mm dia	Nos	6.00	4,950.00	29,700.00
109	50 mm dia	Nos	6.00	3,580.00	21,480.00
	NON RETURN VALVE				
	Supplying and fixing Non return valve of approved make with ISI make including fixing in position, making holes, restoring the damaged portions to original position and applying jointing compound at threaded ends and testing etc. complete as directed				
110	80 mm dia	Nos	4.00	11,640.00	46,560.00
111	65 mm dia	Nos	4.00	11,250.00	45,000.00
112	50 mm dia	Nos	2.00	10,060.00	20,120.00
	GI/MS PUDDLE FLANGE				
	Supplying and fixing of GI/MS Puddle flanges of standard design of following sizes. etc. complete as directed. (for ug sump & oht)				
113	200 mm dia	Nos	2.00	10,700.00	21,400.00
114	150 mm dia	Nos	10.00	7,820.00	78,200.00
115	100 mm dia	Nos	16.00	5,750.00	92,000.00
116	80 mm dia	Nos	10.00	4,720.00	47,200.00
117	65 mm dia	Nos	4.00	4,250.00	17,000.00
118	50 mm dia	Nos	10.00	2,300.00	23,000.00
119	32 mm dia	Nos	4.00	1,900.00	7,600.00
120	25 mm dia	Nos	12.00	1,610.00	19,320.00
	TOTAL OF SECTION - J				8,31,220.00
K	PUMPS				
	RAW WATER PUMPS				
121	Supplying, installing, testing and commissioning approved make monobloc pump made of C.I Construction suitable for building application. The quoted rate shall also include providing necessary safety devices viz. dry run protection, over load relays etc., complete working on 230V, 50Hz, 1Ph, with starter, Duty of Pump : 5.5cum/hr @20.0m Head 1 No. Working + 1 No. Stand By (Raw Water Pumps to WTP)	Set	2.00	63,550.00	1,27,100.00
	Tower-1 Terrace Rain Water Pump				
122	Supplying, installing, testing and commissioning approved make monobloc pump made of C.I Construction suitable for building application. The quoted rate shall also include providing necessary safety devices viz. dry run protection, over load relays etc., complete working on 230V, 50Hz, 1Ph, with starter, Duty of Pump : 6.0cum/hr @30.0m Head 1 No. Working + 1 No. Stand By (Tower-1 Terrace Rain Water Pumps to WTP / External Drain)	Set	1.00	67,100.00	67,100.00

	Tower-2 Terrace Rain Water Pump				
123	Supplying, installing, testing and commissioning approved make monobloc pump made of C.I Construction suitable for building application. The quoted rate shall also include providing necessary safety devices viz. dry run protection, over load relays etc., complete working on 230V, 50Hz, 1Ph, with starter, Duty of Pump : 4.0cum/hr @30.0m Head 1 No. Working + 1 No. Stand By (Tower-2 Terrace Rain Water Pumps to WTP / External Drain)	Set	1.00	57,200.00	57,200.00
	Tower-1 Treated Water Pump				
124	Supplying, installing, testing and commissioning of approved make monoblock water transfer pumps for pumping domestic water from UG Sump to over head tank . The pump shall be coupled to the adequate 10 H.P. electrical motor ,mounted on a common base frame. The motor shall confirm to T.E.F.C. The rate quoted shall be inclusive of The pumps shall be provided with: a) Suitable control panel indoor wall mounting type with change over switch including dry run protection. b) Industrial type electronic automatic liquid level controllers, or float switch for automatic operation. c) Alarm indicating with signal in the event of pumps failure etc., d)GI manifold made out of GI 'C'Class pipe with Collers,Tee and Flange provision for connection of the above items . GI manifold shall be finished with hot dip galvanizing to 100 microns. The pumps shall be suitable for the requirements as mentions below: 10.0LPS @ 45.0m Head 2 Nos. Working + 1 No. Stand By (From Tower 1 Treated Water Sump to O.H.T)	Set	1.00	1,43,000.00	1,43,000.00
	Tower-2 Treated Water Pump				
125	Supplying, installing, testing and commissioning of approved make monoblock water transfer pumps for pumping domestic water from UG Sump to over head tank . The pump shall be coupled to the adequate 10 H.P. electrical motor ,mounted on a common base frame. The motor shall confirm to T.E.F.C. The rate quoted shall be inclusive of The pumps shall be provided with: a) Suitable control panel indoor wall mounting type with change over switch including dry run protection. b) Industrial type electronic automatic liquid level controllers, or float switch for automatic operation. c) Alarm indicating with signal in the event of pumps failure etc., d)GI manifold made out of GI 'C'Class pipe with Collers,Tee and Flange provision for connection of the above items . GI manifold shall be finished with hot dip galvanizing to 100 microns. The pumps shall be suitable for the requirements as mentions below: 8.0LPS @ 45.0m Head 2 Nos. Working + 1 No. Stand By (From Tower 2 Treated Water Sump to O.H.T)	Set	1.00	1,21,000.00	1,21,000.00
	Flushing Water Pump				
126	Supplying, Installing, testing & commissioning approved make variable Hydropneumatic system comprising of Vertical Centrifugal pumps mounted on a common skid with automatic control panel including overload relays, on-off push button switches, phase indicator lamps, dry run protection relays, pressure switches & guages, 1no. FRP pressure vessel of 300 Lts. mounted on the pump complete with cabling between the Pumps, control panel etc., complete. The pumps shall run on 440V, 3ph, 50Hz electric supply. (For Flushing water supply system) Duty of Pump : 6.0 cum/Hr @ 45.0m Head (Each Pump capacity 6.0cum) 3 pump system (2Working+1 Stand By)	Set	1.00	6,00,600.00	6,00,600.00
	Landscaping Water Pump				
127	Supplying, Installing, testing & commissioning approved make variable Hydropneumatic system comprising of Vertical Centrifugal pumps mounted on a common skid with automatic control panel including overload relays, on-off push button switches, phase indicator lamps, dry run protection relays, pressure switches & guages, 1no. FRP pressure vessel of 300 Lts. mounted on the pump complete with cabling between the Pumps, control panel etc., complete. The pumps shall run on 440V, 3ph, 50Hz electric supply. (For landscaping water supply system) Duty of Pump : 7.0 cum/Hr @ 45.0m Head (Each Pump capacity 7.0cum) 2 pump system (1 Working+1 Stand By)	Set	1.00	4,24,600.00	4,24,600.00
	Pumproom Collection Sump pump				
128	Supplying, installing, testing and commissioning approved make submersible pump made of C.I Construction suitable for building application. Thequoted rate shall also include for auto lvl.controller. providing necessary safety devices viz. dry run protection, over load relays etc., complete working on 230V, 50Hz, 1Ph, with starter, Duty of Pump : 1.0 Lps @ 20.0m Head 1 No. Working (From pump room collection tank to drain)	Nos.	2.00	25,850.00	51,700.00
	TOTAL OF SECTION - K				15,92,300.00

L	WATER TREATMENT PLANT				
	Pressure Sand Filter				
129	Supplying, installing, testing and commissioning approved make Raw Water Pressure sand filter with necessary Accessories complete with Frontal piping, valves, suitable supports for mounting pressure gauges, orifice plates. The Filter shall be made of MS with necessary protective coating. The cost of filter shall also include initial charge of underbed & multi-grade filter media to suit the requirement etc., complete as per standards. Max. working Pressure = 3.0kg/sqcm. Max flow rate = 6.0cum/hr.@8hrs cycle	Unit	2.00	1,73,800.00	3,47,600.00
	Activated Carbon Filter				
130	Supplying, installing, testing and commissioning approved make activated carbon filter in MS vessel construction with internals, electrical control system complete with Frontal piping, valves, suitable supports for mounting, pressure gauge, orifice plate etc., complete as per the detailed specification for the efficient working of system. Max. working Pressure = 3.5kg/sqcm. Max flow rate = 6.0cum/hr.@8hrs cycle	Unit	2.00	2,10,100.00	4,20,200.00
	Softner				
131	Supplying, installing, testing and commissioning approved make Water Softener with necessary accessories complete with Frontal piping, valves, suitable supports for mounting pressure gauges, orifice plates. The Softener shall be made of MS with necessary protective coating. The cost of Softener shall also include initial charge of Resin media to suit the requirement etc., complete as per standards. Max. working Pressure = 3.0kg/sqcm. Max flow rate =6.0 cum/hr.@8hrs cycle	Set	2.00	1,73,800.00	3,47,600.00
	Chlorine Dosing System				
132	Supplying, installing, testing and commissioning approved make Electronic dosing system for dosing Chlorine into the Treated water with pre-set dosage based on the the quantity of water. The Dosing system shall consist of Electronic dosing pump with positive displacement type with adjustable Stroke length and frequency. The quoted rate shall also include FRP chemical solution tank, necessary electricals etc., complete to make the operation automatic. Max flow rate =1.0lpm	Set	2.00	27,500.00	55,000.00
	TOTAL OF SECTION - L				11,70,400.00
M	SEWAGE TREATMENT PLANT 80KLD CAPACITY MBBR TYPE				
133	Design, Supply, Erection and commissioning of Prefabricated STP80 KLD Capacity, MBBR Technology including Ultrafiltration system of 5.0Cum/Hr (Treated water used for flushing purpose) with below equipments & tanks as mentioned below. SS bar screen , Raw sewage water pumps, Tubular Air Diffusers for Bio Reactor, Air blower system, Induction motor, Air Grid & feed well, MBBR Reactor Media, Return SludgeE Transfer Pumps, Excess SludgeTransfer Pumps, Filter Press, Wheel Barrow, Filter Feed pumps Pressure Sand filter, Activated Carbon Filter, Treated water Pumps, Chlorine Dosing system, Process Display & Individual units mane borads, flow direction stickers, Electromagnetic Flow Meter, UF Feed pumps, UF Filtration equipments, Tank, membrane, control panel with necessary cables & electrical works, interconnecting pipe & fittings, commissioning chemicals, Fabricated Tanks etc.	KLD	80.00	1,00,000.00	80,00,000.00
	TOTAL OF SECTION - M				80,00,000.00
	TOTAL OF PLUMBING & SANITARY WORKS				5,99,66,700.00

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS								Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT	
I	INTERNAL ELECTRICAL WORKS							
	SUB HEAD I : SANDWICH BUS DUCT/RISING MAINS							
1.0	INDOOR SANDWICH BUSDUCT (TRANSFORMER TO MV PANEL)							
1.1	Supply, Installation, Testing and commissioning of dust & vermin proof 2000A TPN AI 50 KA Indoor Sandwich type IP 52 rated Alu. Bus Duct.The quoted rate shall include ,MS supports, Anchor fasteners, hardware etc.for necessary suspension & supporting from ceiling at suitable intervals etc. as required at site.	Meter	45	35,000	2,500	15,75,000	1,12,500	16,87,500.00
1.2	Supply, Installation of Horizontal/Vertical bend required for the above sandwich Busduct along with additional MS enclosure mentioned above with required fixing accessories & supporting system.	Each	12	10,000	2,000	1,20,000	24,000	1,44,000.00
1.3	Supply, Installation of Flanged end with suitable adopter box required for the above sandwich Busduct for termination at the MV panel side & Transformer side along with copper flexible tape with additional MS enclosure mentioned above with required fixing accessories & supporting system.	Each	4	1,28,000	4,000	5,12,000	16,000	5,28,000.00
2.0	INDOOR SANDWICH BUSDUCT (MV PANEL TO APFC PANEL)							
2.1	Supply, Installation, Testing and commissioning of dust & vermin proof 800A TPN AI 50 KA Indoor Sandwich type IP 52 rated Alu. Bus Duct.The quoted rate shall include ,MS supports, Anchor fasteners, hardware etc.for necessary suspension & supporting from ceiling at suitable intervals etc. as required at site.	Meter	15	28,000	2,000	4,20,000	30,000	4,50,000.00
2.2	Supply, Installation of Horizontal/Vertical bend required for the above sandwich Busduct along with additional MS enclosure mentioned above with required fixing accessories & supporting system.	Each	4	6,000	1,000	24,000	4,000	28,000.00
2.3	Supply, Installation of Flanged end with suitable adopter box required for the above sandwich Busduct for termination at the MV panel side & APFCR panel side along with copper flexible tape with additional MS enclosure mentioned above with required fixing accessories & supporting system.	Each	2	60,000	2,500	1,20,000	5,000	1,25,000.00
3.0	RISING MAIN -1&1A,2&2A							-
3.1	Supply, Installation, Testing & Commissioning of Sandwich type 630 A Aluminium 4 POLE 35 ka indoor Rising mains,twin 25&3mm Cu earth bar with plug in Tap off box as per requirement.Every floor should have minimum two nos tap-off provisions The tap off boxes 's should have sufficient space and clearance with extended busbars to terminate copper flexible cables with suitable supports. The scope includes providing necessary supports, Anchor fasteners,Flair up units with busbars,to connect with isolation panels, floor spring hangers suitable end cover at Top of the RM with required hardware etc. for necessary suspension & supporting from ceiling / wall/RCC for the busduct.(IP 52).							
a	Horizontal Run	Mtrs.	200	18,000	2,500	36,00,000	5,00,000	41,00,000.00
b	Horizontal Bend	Nos.	8	6,500	1,000	52,000	8,000	60,000.00
c	Vertical Bend	Nos.	8	6,500	1,000	52,000	8,000	60,000.00
d	Elbow	Nos.	4	9,500	1,000	38,000	4,000	42,000.00
e	End Termination with flexible Bellows	Nos.	4	30,000	2,000	1,20,000	8,000	1,28,000.00
f	Vertical Straight run	Mtrs.	140	18,500	2,500	25,90,000	3,50,000	29,40,000.00
g	Plug in type tap off box with 250 A 35KA TPN MCCB with Fixed Thermal Over Load and Fixed Magnetic Short Circuit Protection with RHOM.	Nos.	40	36,500	1,200	14,60,000	48,000	15,08,000.00
h	Plug in type tap off box with 63 A 35KA TPN MCCB with Fixed Thermal Over Load and Fixed Magnetic Short Circuit Protection with RHOM.	Nos.	18	25,000	1,200	4,50,000	21,600	4,71,600.00
	TOTAL OF SUB HEAD I: SANDWICH BUS DUCTS/RISING MAINS					1,11,33,000	11,39,100	1,22,72,100.00
	Sub Head II : LT PANELS							
	Supply, Installation, Testing and Commissioning of following LT panels as per Specification and Particular requirement							
1.0	Main MV panel Indoor (As per SLD)							
	EB Incomer-1&2							
	1 No 2000 Amps 50 ka FP ACB EDO with micro processor based LSIG Protection.ACB's shall have the communication option as per Technical specifications							
	Metering:							
	DIGITAL type flush mounting Load manager / Controller with LED display for Current, Voltage, KW,KWhr, Power factor, Frequency, THD V & I with RS 485 communication port and Sensing, indication & tripping arrangement for demand control CI 1.0 Accuracy and 3 Sets of CT 2000 A/5 A ,15 VA CI 1.0 accuracy.							
	RYB/ON/OFF/Trip/Healthy 22 mm LED Indicator.							
	Protection							
	1set of Surge protection device 'B'							
	1 No. - Integral Microprocessor based Over load and Restricted Earth Fault Relay							
	3 Nos. 2000 / 5A CI 1.0 15VA CT.(For APFCR panel)							
	DG Incomer - 1No.							
	1No. 400A,4P Micro processor based MCCB, 50KA							
	Metering:							
	DIGITAL type flush mounting Load manager with LED display for Current, Voltage, KW,KWhr, Power factor, Frequency, THD V & I with RS 485 communication port and Sensing, indication & tripping arrangement for demand control CI 1.0 Accuracy and 3 Sets of CT 400 A/5 or 1 A,5 VA CI 1.0 accuracy.							
	1No. Surge protection device 'C'							
	BUSCOUPLER - 1No.							
	1 No 2000A 4P , Electrically operated, Draw out type ACB with No protections with necessary accessories							
	Bus Bars							
	1 set of 2000 A TPN 50KA Aluminium Bus Bar							
	Outgoing section							
	1 No. 800 Amps 50 ka TPN MCCB with micro processor based LSIG Protection with suitable CT & Digital Ammeter.							
	5 Nos 630 A TPN 50 ka MCCB with suitable CT& Dual Source Energy Meter.							
	2 Nos 400 A TPN 50 ka MCCB set with suitable CT& Dual Source Energy Meter.							
	2 Nos 250 A TPN 50 ka MCCB set with suitable CT& Dual Source Energy Meter.							
	1 Nos 200 A TPN 50 ka MCCB set with suitable CT& Dual Source Energy Meter.	set	1	21,27,500	35,000	21,27,500	35,000	21,62,500.00
2.0	Auto Power Factor Correction Capacitor panel (APFCR). 425KVAR							
	Incomer							
	1 No 800 A, 35ka,TPN, MCCB.							
	APFCR Relay with suitable CT's							
	Set of LED lamps for indications and trips with MCB control							
	BUSBAR							
	1 set of 800 A TPN Aluminium Bus Bar							
	Outgoings							
	3 sets of 100 KVAR MPP Gas filled type capacitors, 550 Volts, 'ISi' marked 7 % Detuned reactors ,250 A, TP, 35KA MCCB with rotary handle,Current limiting type TP contactors suitable for capacitor duty,ON/OFF indicating lamp complete with control wiring from APFCR							
	2 sets of 50 KVAR MPP Gas filled type capacitors, 550 Volts, 'ISi' marked 7 % Detuned reactors ,125 A, TP, 35KA MCCB with rotary handle,Current limiting type TP contactors suitable for capacitor duty,ON/OFF indicating lamps complete with control wiring from APFCR							
	2 set of 10 KVAR MPP Gas filled type capacitors, 550 Volts, 'ISi' marked 7 % Detuned reactors , 63 A, TP, 35KA MCCB with rotary handle,Current limiting type TP contactors suitable for capacitor duty,ON/OFF indicating lamp complete with control wiring from APFCR							
	1 set of 5 KVAR MPP Gas filled type capacitors, 550 Volts, 'ISi' marked 7 % Detuned reactors ,32A, TP, 35KA MCCB with rotary handle,Current limiting type TP contactors suitable for capacitor duty,ON/OFF indicating lamp complete with control wiring from APFCR							
	Micro Processor based Automatic Power Power Factor relay 8 stages with Digital display for Amp,Voltage,PF and KVAR as per specifications.							
	Auto manual selector switch with facility for manual control capacitor bank in the event automatic operation not functioning.							
	Fan (with thermostat) for cooling of panel	set	1	11,50,000	15,000	11,50,000	15,000	11,65,000.00
3.0	MLSB PANEL							
	Incomer							
	1 Nos 200 Amps TPN 35 ka MCCB							

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS									Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT		
	METER								
	Digital Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomers								
	BUSBAR								
	1 set of 200 A TPN,35KA Aluminium Bus Bar								
	Outgoings								
	10 nos 63 Amps TPN 35 KA MCCB	set	1	3,25,000	12,000	3,25,000	12,000		3,37,000.00
4.0	MPSB PANEL								
	Incomer								
	1 Nos 250 Amps TPN 35 KA MCCB								
	METER								
	Digital Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomers								
	BUSBAR								
	1 set of 250 A TPN,35KA Aluminium Bus Bar								
	Outgoings								
	2 nos. 100 Amps TPN 35 KA MCCB								
	7 nos. 63 Amps TPN 35 KA MCCB	set	1	3,25,000	12,000	3,25,000	12,000		3,37,000.00
5.0	STP PANEL								
	Incomer								
	1 Nos 100 Amps TPN 25 KA MCCB								
	METER								
	Digital Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomers								
	BUSBAR								
	1 set of 100 A TPN 25KA Aluminium Bus Bar								
	Outgoings								
	4 nos 63 Amps TPN MCCB								
	4 nos 32 Amps TPN MCCB	set	1	2,70,000	7,500	2,70,000	7,500		2,77,500.00
6.0	WTP PANEL								
	Incomer								
	1 Nos 63 Amps TPN 25 KA MCCB								
	METER								
	Digital Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomers								
	BUSBAR								
	1 set of 63 A TPN 25KA Aluminium Bus Bar								
	Outgoings								
	3 nos 32 Amps TPN MCCB								
	3 nos 16 Amps TPN MCCB	set	1	2,45,000	7,500	2,45,000	7,500		2,52,500.00
7.0	TYPICAL FLOOR PANEL								
	Incomer								
	250A,4P,Change Over Switch - 1No.								
	METER								
	Digital Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomers - 2set								
	BUSBAR								
	1 set of 250 A TPN 25KA Aluminium Bus Bar								
	Outgoings								
	10 nos 63 Amps TPN MCCB with Multifunction meter with RS 485 port	set	18	5,50,000	9,000	99,00,000	1,62,000		1,00,62,000.00
8.0	TERRACE FLOOR PANEL								
	Incomer								
	250A,4P, Auto Change over switch - 1No.								
	METER								
	Digital Dual source Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomers - 2set								
	BUSBAR								
	1 set of 250 A TPN 25KA Aluminium Bus Bar								
	Outgoings								
	10 nos 63 Amps TPN MCCB with Multifunction meter with RS 485 port	set	2	5,45,000	9,000	10,90,000	18,000		11,08,000.00
9.0	TYPICAL FLOOR DG PANEL								
	Incomer								
	40A,4P,MCB - 1No.								
	4Pole Contactor - 1No.								
	METER								
	Digital Multifunction meter - 1 set								
	INDICATION								
	RYB 22 mm dia LED Lamp & On/Off/Trip Indication for incomer - 1set								
	BUSBAR								
	1 set of 63A TPN 16KA Aluminium Bus Bar								
	Outgoings								
	12 nos 6A DP MCB with Energy meter with RS 485 port	set	18	1,47,000	2,500	26,46,000	45,000		26,91,000.00
10.0	ISOLATORS								
a	63A,4P MCB with Non-Metallic enclosure	set	2	4,500	800	9,000	1,600		10,600.00
b	63A,TPN,MCCB with Non-metallic enclosure (Lift)	set	4	15,000	1,250	60,000	5,000		65,000.00
c	10A,DP,MCB with Non-Metallic enclosure (Staircase light control)	set	4	2,500	500	10,000	2,000		12,000.00
	TOTAL OF SUB HEAD II :LT PANELS					1,81,57,500	3,22,600		1,84,80,100.00
	SUB HEAD III - LT CABLES & CABLE TRAY								
1.0	SUPPLYING & LAYING CABLE								
	Supply,laying,testing and commissioning of following sizes of XLPE Insulated, cores laid up, PVC tape/Extruded Inner sheathed,galvanised steel strip armoured, Extruded PVC Type ST2 Sheathed, Aluminium / Copper conductor 1.1 kv grade cable as per IS 7098(Part 1) 1988 on wall/ceiling/ cable tray / masonry duct/pipes with suitable MS Clamps wherever required as per specification								
a	3.5 Core 185 sq.mm XLPE,Al,Ar.	Meter	20	1,050	180	21,000	3,600		24,600.00
b	3.5 Core 150 sq.mm XLPE,Al,Ar.	Meter	80	850	140	68,000	11,200		79,200.00
c	3.5 Core 120 sq.mm XLPE,Al,Ar.	Meter	120	700	120	84,000	14,400		98,400.00
d	3.5 Core 50 sq.mm XLPE,Al,Ar.	Meter	50	315	100	15,750	5,000		20,750.00
e	3.5 Core 25 sq.mm XLPE,Al,Ar.	Meter	200	200	100	40,000	20,000		60,000.00
f	4 Core 16 sq.mm XLPE,Al,Ar.	Meter	380	160	80	60,800	30,400		91,200.00
g	3.5 Core 35 sq.mm PVC CU Ar.	Meter	440	1,600	90	7,04,000	39,600		7,43,600.00
h	4 Core 10 sq.mm PVC CU Flexible	Meter	75	580	75	43,500	5,625		49,125.00
i	4 Core 10 sq.mm PVC CU Ar.	Meter	60	500	75	30,000	4,500		34,500.00
j	4 Core 6 sq.mm PVC CU Ar.	Meter	305	390	50	1,18,950	15,250		1,34,200.00
k	4 Core 4 sq.mm PVC CU Ar.	Meter	45	270	45	12,150	2,025		14,175.00
l	4 Core 2.5 sq.mm PVC CU Ar.	Meter	161	180	40	28,980	6,440		35,420.00
m	19 Core 2.5 sqmm Copper Ar. Cable (Tr - RTCC)	Meter	40	715	125	28,600	5,000		33,600.00
n	2 Core 1.5 sq.mm PVC CU Ar.	Meter	300	80	20	24,000	6,000		30,000.00
2.0	END TERMINATION								
	Supplying and making end termination with double compression gland and aluminium/copper lugs for following size of PVC insulated and PVC sheathed / XLPE aluminium/copper conductor cable of 1.1 KV grade as required.								
a	3.5 Core 185 sq.mm XLPE,Al,Ar.	Nos	2	1,800	850	3,600	1,700		5,300.00
b	3.5 Core 150 sq.mm XLPE,Al,Ar.	Nos	4	1,600	750	6,400	3,000		9,400.00

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS								Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT	
c	3.5 Core 120 sq.mm XLPE AL Ar.	Nos	8	1,400	500	11,200	4,000	15,200.00
d	3.5 Core 50 sq.mm XLPE AL Ar.	Nos	2	800	200	1,600	400	2,000.00
e	3.5 Core 25 sq.mm XLPE AL Ar.	Nos	4	500	150	2,000	600	2,600.00
f	4 Core 16 sq.mm XLPE AL Ar.	Nos	20	400	100	8,000	2,000	10,000.00
g	3.5 Core 35 sq.mm PVC Cu Ar.	Nos	88	900	150	79,200	13,200	92,400.00
h	4 Core 10 sq.mm PVC Cu flexible	Nos	10	340	100	3,400	1,000	4,400.00
i	4 Core 10 sq.mm PVC Cu Ar.	Nos	8	330	90	2,640	720	3,360.00
j	4 Core 6 sq.mm PVC Cu Ar.	Nos	54	300	70	16,200	3,780	19,980.00
k	4 Core 4 sq.mm PVC Cu Ar.	Nos	6	270	60	1,620	360	1,980.00
l	4 Core 2.5 sq.mm PVC Cu Ar	Nos	14	200	50	2,800	700	3,500.00
m	19 Core 2.5 sq.mm Copper Ar. Cable (Tr - RTCC)	Nos	4	500	200	2,000	800	2,800.00
n	2 Core 1.5 sq.mm PVC Cu Ar	Nos	40	100	30	4,000	1,200	5,200.00
3.0	LADDER TYPE CABLE TRAY							
	Supply and erection of following sizes GI Ladder Type Tray made out of sheet steel folded to required shape and adequate no. supports. The cable Tray shall be suspended from ceiling fixed to walls/columns etc. using adequate size supports, angles/channels with fasteners and anchor bolts. The Tray shall have suitable cable clamping arrangements.							
a	1000 mm wide Runners 25 x 100 x 25 x 2.5 mm Rungs 20 x 30 x 20 x 3 mm 250 mm C/C Suspenders 2 Nos. 32 x 32 x 5 mm GI angle. or 12 mm MS Threaded Rod 600 mm C/C	Meter	100	1150	350	1,15,000	35,000	1,50,000.00
b	750 mm wide Runners 25 x 100 x 25 x 2.5 mm Rungs 20 x 30 x 20 x 3 mm 250 mm C/C Suspenders 2 Nos. 32 x 32 x 5 mm GI angle. or 12 mm MS Threaded Rod 600 mm C/C	Meter	150	1000	300	1,50,000	45,000	1,95,000.00
c	600 mm wide Runners 25 x 100 x 25 x 2.5 mm Rungs 20 x 30 x 20 x 3 mm 250 mm C/C Suspenders 2 Nos. 32 x 32 x 5 mm GI angle. or 12 mm MS Threaded Rod 600 mm C/C	Meter	200	850	250	1,70,000	50,000	2,20,000.00
d	450 mm wide Runners 25 x 100 x 25 x 2.5 mm Rungs 20 x 30 x 20 x 3 mm 250 mm C/C Suspenders 2 Nos. 25 x 25 x 4 mm GI angle. or 8 mm MS Threaded Rod 1000 mm C/C	Meter	200	750	250	1,50,000	50,000	2,00,000.00
4.0	Supply, erection, testing & commissioning of following sizes of GI covered Perforated type Sheet Metal cable trays and perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with M.S. suspenders including bolts & nuts, anchor fastners. painting suspenders etc including painting of suspenders etc as per detailed specification.							
a	300 mm width X 50 mm depth X 1.6 mm thickness	Meter	200	750	150	1,50,000	30,000	1,80,000.00
b	150 mm width X 50 mm depth X 1.6 mm thickness	Meter	300	500	100	1,50,000	30,000	1,80,000.00
	TOTAL OF SUB HEAD III : LT CABLES & CABLE TRAY					23,09,390	4,42,500	27,51,890.00
	SUB HEAD IV-DISTRIBUTION BOARDS & INVERTER							
1.0	VTN MCB DISTRIBUTION BOARDS							
	Supplying and fixing following way prewired Vertical TP&N MCB distribution board made of steel sheet for surface/ recess mounted complete with loose wire box, terminal connectors for all incoming and outgoing circuits, duly prewired with suitable size FRLS PVC insulated copper conductor up to terminal blocks, tinned copper bus bar, neutral link, earth bar, din bar, detachable gland plate, interconnections, powder coated painted including earthing etc. as required. IP-43 with metal door							
a	6 WAY External LDB							
	Incomer							
	63 A TPN MCB							
	0-24Hrs Digital timer to control the contactor							
	Out Goings							
	3 Nos 16A TP MCBs 10KA 'C' Curve with contactor.							
	3 Nos 16A TP MCBs 10KA 'C' Curve.	Each	1	28,000	2,500	28,000	2,500	30,500.00
2.0	TPN MCB DISTRIBUTION BOARDS							
	Supplying and fixing following way prewired MCB distribution board of steel sheet on surface/ recess mounted complete with loose wire box, terminal connectors for all incoming and outgoing circuits, duly prewired with suitable size FRLS PVC insulated copper conductor up to terminal blocks, tinned copper bus bar, neutral link, earth bar, din bar, detachable gland plate, interconnections, powder painted including earthing etc. as required. IP-43 with metal door							
a	8 WAY TPN MCB LDB - (Ground/First floor/Typ floor)							
	Incomer							
	63 A TPN MCB							
	Sub Incomer							
	3Nos of 25A DP RCCB 30mA							
	Out Goings							
	24 Nos 6A SP MCBs	Each	6	20,000	2,000	1,20,000	12,000	1,32,000.00
b	6 WAY TPN MCB LDB - (Terrace/Recreation/Gym)							
	Incomer							
	63 A TPN MCB							
	Sub Incomer							
	3Nos of 25A DP RCCB 30mA							
	Out Goings							
	18 Nos 6A SP MCBs	Each	4	16,500	2,000	66,000	8,000	74,000.00
c	6 WAY TPN MCB PDB - (Ground/First floor/Terrace)							
	Incomer							
	63 A TPN MCB							
	Sub Incomer							
	3Nos of 40A DP RCCB 30mA							
	Out Goings							
	18 Nos 16A SP MCBs	Each	6	16,500	2,000	99,000	12,000	1,11,000.00
d	8 WAY TPN MCB EMERGENCY LDB							
	Incomer							
	63 A TPN MCB							
	Sub Incomer							
	3Nos of 40A DP RCCB 30mA							
	Out Goings							
	24 Nos 10A SP MCBs	Each	1	20,000	2,000	20,000	2,000	22,000.00
e	6 WAY TPN MCB 8 Segment LPDB (Typical for 3/2BHK)							
	Incomer							
	63A TPN MCB + 3A DP MCB - 1No.							
	ACCL 3A in DG and 25A in EB - 1No.							
	Sub Incomer							
	2Nos of 40A DP RCCB 30mA							
	1No. of 25A DP RCCB 30mA							
	Out Goings							
	18 Nos 6A/16A/25A SP MCBs	Each	126	22,500	2,500	28,35,000	3,15,000	31,50,000.00
3.0	INVERTER-10 KVA							
a	Supply/Installation of 10KVA UPS Three phase input/ output with 30min.battery backup (As per specification)	Set	1	1,70,000	5,000	1,70,000	5,000	1,75,000.00
	TOTAL OF SUB HEAD IV-DISTRIBUTION BOARDS & UPS					33,38,000	3,56,500	36,94,500.00
	SUB HEAD V - LIGHT & POWER POINT WIRING							
1.0	PRIMARY LIGHT POINT WITH SWITCH							
	Wiring for Light Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FR-LSH PVC Insulated Stranded copper conductor single core cable with recessed ISI PVC conduit including providing and fixing suitable Galvanised / Cadmium plated MS box, 6A one-Way modular Switch, modular plate and making necessary connections, earthing etc. complete as per specification.	Points	2800	1,000	500	28,00,000	14,00,000	42,00,000.00
2.0	SECONDARY LIGHT POINT							

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS								Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT	
	Wiring for additional /primary/secondary light points with out switch, switch box & modular plate (Group Controlled by switch or MCB in MCBDB) with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FR-LSH PVC insulated stranded copper conductor Single core cable with recessed ISI PVC conduit making necessary connections, earthing etc. complete as per specification etc.,	Points	705	300	150	2,11,500	1,05,750	3,17,250.00
3.0	CEILING FAN POINT							
	Wiring for Fan Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FR-LSH PVC insulated multistranded copper wire in recessed ISI PVC conduit with fan box including providing and fixing suitable Galvanised / Cadmium plated MS box, 6A one-Way modular Switch, 100 Watt modular step Regulator(2 M), modular plate and making necessary connections, earthing etc. complete as per specification.	Points	580	1,300	700	7,54,000	4,06,000	11,60,000.00
4.0	2WAY LIGHT POINT							
	Wiring for Two way Light Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FRLS PVC insulated multistranded copper wire in recessed ISIPVC conduit including providing and fixing 2 Nos suitable Galvanised / Cadmium plated MS box, 6A Two-Way modular Switch, modular plate and making necessary connections, earthing etc. complete as per specification.	Points	306	1,700	650	5,20,200	1,96,900	7,19,100.00
5.0	EXHAUST FAN POINT							
	Wiring for Exhaust Fan Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FR-LSH PVC insulated Stranded copper conductor single core cable in recessed ISIPVC conduit including providing and fixing suitable Galvanised / Cadmium plated MS box, 6A one-Way modular Switch, modular plate and making necessary connections, earthing etc. complete as per specification.	points	444	1,200	550	5,32,800	2,44,200	7,77,000.00
6.0	CALLING BELL POINT							
	Wiring for Calling Bell Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FRLS PVC insulated Stranded copper conductor single core cable in recessed ISI PVC conduit including providing and fixing suitable Galvanised / Cadmium plated MS box, 6A Bell push Switch, modular plate and making necessary connections etc. complete as per specification.	points	126	1,100	500	1,38,600	63,000	2,01,600.00
7.0	DEPENDENT 6 A SPIN SOCKET OUTLET							
	Supplying and fixing of 6 A one-way modular Switch, 6 A, 5 pin, modular shuttered socket outlet and making connections, earthing etc., complete as per specification. / Switch Board Plug	Nos.	800	200	150	1,60,000	1,20,000	2,80,000.00
8.0	INDEPENDENT 6 A SPIN SOCKET OUTLET							
	Wiring for Independent 6 A socket outlet with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FR-LSH PVC insulated multistranded copper wire in recessed 2mm thick PVC conduit including providing and fixing suitable Galvanised / Cadmium plated MS box, 6A one-Way modular Switch, 6 A, 5 pin modular Socket outlet, modular plate and making necessary connections, earthing etc. complete as per specification.	Points	954	1,300	650	12,40,200	6,20,100	18,60,300.00
9.0	INDEPENDENT 2 X 6 A SPIN SOCKET OUTLET FOR TV POINT							
	Wiring for Independent Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). FR-LSH PVC insulated multistranded copper wire in recessed ISIPVC conduit including Providing and fixing suitable Galvanised / Cadmium plated MS box, 2 X 6A one-Way modular Switch, 2 X 6 A, 5 pin modular Socket outlet, modular plate and making necessary connections, earthing etc. complete as per specification.	Points	432	1,726	700	7,45,632	3,02,400	10,48,032.00
10.0	16A SOCKET OUTLET- CONDUIT & WIRING							
	Wiring for 16A socket outlet with 2 x 2.5 sq. mm. along with 1 x 2.5 sq. mm (E). FR-LSH PVC insulated multistranded copper wire in recessed ISIPVC conduit with necessary accessories etc. complete as per specification.	Mtrs	3000	150	40	4,50,000	1,20,000	5,70,000.00
11.0	16A SOCKET OUTLET- SWITCH BOX							
	16/6A 5 pin Switched Socket with Front plate and flush mounting MS box with necessary items etc.,	Nos.	400	855	150	3,42,000	60,000	4,02,000.00
12.0	GEYSER POINT - CONDUIT & WIRING							
	Wiring for Geyser point with 2 x 2.5 sq. mm. along with 1 x 2.5 sq. mm (E). FRLS PVC insulated multistranded copper wire in recessed ISIPVC conduit with necessary accessories etc. complete as per specification.	Mtrs	6000	150	40	9,00,000	2,40,000	11,40,000.00
13.0	GEYSER POINT (SWITCH & SOCKET BOX)							
	16A Switch with indicator, front plate & 2module flush mounting box with necessary accessories etc.,	Nos.	306	538	150	1,64,628	45,900	2,10,528.00
	16A/6A 5 pin Socket with front plate & 2module flush mounting box with necessary accessories etc.,	Nos.	306	656	150	2,00,736	45,900	2,46,636.00
14.0	AC POINT (CONDUIT & WIRING)							
	Wiring for AC point with 2 x 4 sq. mm. along with 1 x 2.5 sq. mm (E). FR-LSH PVC insulated multistranded copper wire in recessed ISIPVC conduit with necessary accessories etc.,	Mtrs	9000	152	50	13,68,000	4,50,000	18,18,000.00
15.0	AC POINT (SWITCH & SOCKET)							
	Supply and fixing of suitable Galvanised / Cadmium plated 2module MS box with modular plate with 20A DP switch with indicator etc.,	Nos.	442	600	150	2,65,200	66,300	3,31,500.00
	Supply and fixing of suitable Galvanised / Cadmium plated 2module MS box with modular plate with 20A 3pin socket etc.,	Nos.	442	600	150	2,65,200	66,300	3,31,500.00
16.0	DB TO PRIMARY LIGHTING SWITCH BOX							
	Wiring for Primary Lighting switch box with 2 x 2.5 sq. mm. along with 1 x 2.5 sq. mm (E). FR-LSH PVC insulated multistranded copper wire in recessed ISI PVC conduit with necessary accessories etc.,	Mtrs.	15000	150	40	22,50,000	6,00,000	28,50,000.00
17.0	PRIMARY TO SECONDARY LIGHTING SWITCH BOX							
	Wiring for Secondary Lighting switch box with 3 x 2.5 sq. mm FR-LSH PVC insulated multistranded copper wire in recessed ISI PVC conduit with necessary accessories etc.,	Mtrs.	8000	150	40	12,00,000	3,20,000	15,20,000.00
	Supplying and Fixing 6 MODULE cadmium plated / galvanised M.S Switch box in recess with Hylum Sheet cover in front as per specifications (Telephone/TV Junction box)	Nos.	126	250	150	31,500	18,900	50,400.00
18.0	CONDUIT & ACCESSORIES							
	ISI PVC CONDUIT-HEAVY DUTY FOR CEILING, MEDIUM DUTY FOR WALL							
a	25mm Dia PVC Conduit	Mtrs	5000	40	20	2,00,000	1,00,000	3,00,000.00
b	32mm Dia PVC flexible hose	Mtrs	400	30	25	12,000	10,000	22,000.00
19.0	MAIN WIRING (Floor/DG panel to Flat DB)							
	Wiring for Lighting / Power circuit/Submain etc along with earth wire with following sizes of ZH -FRLS PVC insulated multistranded copper wire in already laid conduit as per specification.							
a	4 X 16 sq. mm + 2 X 6 sq. mm Earth wire in Existing conduit. (3/2BHK-DB (EB Incomer))	Mtrs	3200	1,300	70	41,60,000	2,24,000	43,84,000.00
b	2 X 2.5 sq. mm ZHFRLS Cu wire in Existing conduit. (3/2BHK-DB (DG Incomer))	Mtrs	3200	60	20	1,92,000	64,000	2,56,000.00
20.0	STAIRCASE LIGHT POINTS							
	Wiring for Staircase Light Point with 2 x 1.5 sq. mm. along with 1 x 1.5 sq. mm (E). ZH-FRLS PVC insulated Stranded copper conductor single core cable in recessed ISI PVC conduit.	Points	96	650	400	62,400	38,400	1,00,800.00
21.0	LIFT WELL LIGHT, SOCKET, BULK HEAD							
	Wiring for Lift Well Light Point with 2 x 1.5 sq. mm. + 1 x 1.5 (E) sq. mm. ZH-FRLS PVC insulated Multi Stranded copper wire laid in surface with 1.6mm thick MS Conduit including providing and fixing Galvanised / Cadmium plated MS box, 2 nos 6A Single Way modular Switch, 1 no 6A 5 Pin Shuttered Socket outlet, modular plate in surface, Lamp and making necessary connections etc. complete as per specification.	Points	52	1,700	800	88,400	41,600	1,30,000.00
	TOTAL OF SUB HEAD V : LIGHT & POWER POINT WIRING					1,92,64,996	59,71,650	2,52,26,646.00
	SUB HEAD VI : LOW CURRENT WIRING							
1	Supply & Drawing high density polyethylene insulated, solid annealed high conductivity 0.5 mm dia bare copper, following pair, polyester taped and sheathed with high oxygen index Fire Retardant PVC compound, generally conforming to ITD specifications in existing recessed PVC conduit making connections etc as required.							
a	2 Pair	Mtrs.	4000	12	6	48,000	24,000	72,000.00
b	10 Pair	Mtrs.	700	100	40	70,000	28,000	98,000.00
c	100 Pair	Mtrs.	150	600	120	90,000	18,000	1,08,000.00
d	200 Pair	Mtrs.	150	1,200	175	1,80,000	26,250	2,06,250.00
	KRONE CONNECTOR							
2	Supplying and fixing of following pair krone type PVC/MS Powder Coated telephone junction box in surface with hinged cover and screwed type tag strips i/c testing of all telephone cable numbering and making connection as reqd.							
a	10 Pair	Nos	20	750	100	15,000	2,000	17,000.00
b	100 Pair	Nos	2	2,500	500	5,000	1,000	6,000.00
c	200 Pair	Nos	1	4,000	900	4,000	900	4,900.00
3	TELEPHONE SOCKET							
a	Supply and fixing of RJ 11 Telephone socket with front plate and back boxes with necessary accessories etc.,	Nos	150	150	100	22,500	15,000	37,500.00
4	EPABX							
a	100 line extension Digital EPABX	Nos	2	19,000	2,000	38,000	4,000	42,000.00

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS								Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT	
5	RG 6 TV COAXIAL CABLE & SOCKET							
a	Supply & Laying of RG 6 TV foam insulated copper conductor double screened with aluminium foil and aluminium alloy braiding Coaxial Cable in the existing PVC conduit complete as required	Mtrs.	9000	15	8	1,35,000	72,000	2,07,000.00
b	Supply and fixing of Co-axial TV socket with front plate and back boxes with necessary accessories etc.,	Nos	450	150	100	67,500	45,000	1,12,500.00
6	CAT 6 CABLE & SOCKET							
a	Supply & Laying of CAT 6 cable in the already laid PVC conduit complete as required	Mtrs.	QRO	15	8			
b	Supply and fixing of CAT6 socket with front plate and back boxes with all necessary accessories including.	Nos	QRO	500	150			
7	SUPPLYING AND FIXING OF FRLS PVC CONDUIT							
	Supplying and fixing of following sizes of ISI FRLS PVC conduit along with the accessories in recess including cutting the wall and making good the same in case of recessed conduit as required:							
a	20mm DIA FRLS PVC CONDUITS	Mtrs.	8000	45	20	3,60,000	1,60,000	5,20,000.00
b	25mm DIA FRLS PVC CONDUITS	Mtrs.	4000	60	25	2,40,000	1,00,000	3,40,000.00
8	SUPPLYING AND FIXING OF MS CONDUIT							
	Supplying and fixing of following sizes of MS conduit along with the required accessories in surface.							
a	25mm	Mtrs.	500	130	25	65,000	12,500	77,500.00
b	32mm	Mtrs.	300	165	35	49,500	10,500	60,000.00
c	40mm	Mtrs.	150	200	40	30,000	6,000	36,000.00
	Total for SH VI : LOW CURRENT WIRING:					14,19,500	5,25,150	19,44,650.00
	TOTAL FOR INTERNAL ELECTRICAL WORKS					5,56,12,388	87,87,500	6,43,99,886.00
II	EXTERNAL ELECTRICAL WORKS							
	SUB HEAD I - HT & TRANSFORMERS							
1.0	OUTDOOR 11 KV EB RMG- As per EB standards & Specifications							
	Supply, Installation, Testing and Commissioning of 11 KV, 26.3 KA, 630 A, Out door Type 3 Panel Ring Main Unit (RMG) Consisting of 2 nos load Break Switch with Earth Switch and 1 no Vacuum Circuit Breaker with 630 A Copper Busbar, 11 KV Metering Cubicle as per Local EB Specifications and obtaining approval from EB .	Each	1	9,50,000	25,000	9,50,000	25,000	9,75,000.00
2.0	11KV HT Single VCB panel (Outdoor)							
	11 KV, 26.3 KA 630 A, Outdoor Type HT Single Panel Vacuum Circuit Breaker panel with 630 A Copper Busbar, surge diverter, 60/30A Dual CT with Trivector meter and with IDMT earth fault and over current relay.	Each	1	5,25,000	15,000	5,25,000	15,000	5,40,000.00
3.0	11KV HT Three VCB panel (Indoor)							
	11 KV, 26.3 KA 630 A, Indoor Type HT Three Panel Vacuum Circuit Breaker panel with 1No. 630A VCB as incomer and 2nos outgoing with 630A VCB breaker with 630 A Copper Busbar, surge diverter, 60/30A Dual CT with Trivector meter and with Instantaneous earth fault and over current relay for transformer protection.	Each	1	13,35,000	30,000	13,35,000	30,000	13,65,000.00
4.0	Dry type Transformer							
	Design, manufacture, shop testing at works, supply, assembling, installation, precommissioning testing at site and commissioning of 1250 KVA , AN, 11 KV /0.433 kV, 50 Hz, Dyn 11, Cast Resin Dry type transformer, copper winding, degree of protection IP - 52 complete with H.V cable box with disconnecting chamber having inspection cover suitable to receive 11KV (E), suitable 3C HT cable and LV bus duct termination, Winding temperature indicator, ON Load tap changer with RTCC Panel with AVR (+5% to -15% in steps of 1.5 %), one additional neutral terminal shall be brought out with one No. 1000/5A Class 1.0 SP 10 15VA CT in neutral to earth connection for Restricted Earth Fault protection of transformer; standard mounting accessories, fittings, etc.,	Each	2	38,50,000	25,000	77,00,000	50,000	77,50,000.00
5.0	HT CABLES							
	Note: The contractor shall verify size of HT cable to be used from EB/CEIG and intimate the same for approval.							
5.1	Supply & Laying of following sizes of XLPE Insulated, cores laid up, PVC tape/Extruded Inner sheathed, galvanised steel strip armoured, Extruded PVC Type ST2 Sheathed, Aluminium conductor 11 kv grade cable as per IS 7098 (Part 1) 1988 .							
a	11 KV 3 Core 240 sq. mm	Meter	200	1,900	500	3,80,000	1,00,000	4,80,000.00
5.2	Supply and making cable end Termination with Heat Shrinkable jointing kit complete with all accessories including lugs, for the following sizes XLPE Armored Aluminium conductor cable of 33 KV Grade as required.							
a	11 KV 3 Core 240 sq. mm	Set	10	18,000	8,000	1,80,000	80,000	2,60,000.00
6.0	TRENCH EXCAVATION & PROTECTIVE COVERING							
6.1	Excavation of trench up to depth of 1 meter and back filling of trench with excavated earth, light ramming to make the road/ground surface as it was, including providing 150 mm sand /quary dust cushioning, protective covering of cable with RCC "V" shaped cable turf or RCC Tiles as protective covering of cable . (HT cable)	Meter	180	850	350	1,53,000	63,000	2,16,000.00
7.0	RCC HUMPE PIPE							
7.1	Supplying and Laying of NP-2 Grade RCC Hume pipes of following sizes direct in ground at a depth of 1 meter below ground including excavation and refilling the trench etc. as required.							
a	200 mm	Meter	30	800	350	24,000	10,500	34,500.00
	TOTAL OF SUB HEAD I: HT & TRANSFORMERS					1,12,47,000	3,73,500	1,16,20,500.00
	SUB HEAD II: LT CABLES							
1.0	SUPPLYING & LAYING CABLE							
	Supply, laying, testing and commissioning of following sizes of XLPE Insulated, cores laid up, PVC tape/Extruded Inner sheathed, galvanised steel strip armoured, Extruded PVC Type ST2 Sheathed, Aluminium / Copper conductor 1.1 kv grade cable as per IS 7098 (Part 1) 1988 on wall/ceiling/ cable tray / masonry duct/pipes with suitable MS Clamps wherever required as per specification.							
a	4 Core 6 sq. mm XLPE ALAr	Meter	450	100	50	45,000	22,500	67,500.00
2.0	END TERMINATION							
	Supplying and making end termination with double compression gland and aluminium/copper lugs for following size of PVC insulated and PVC sheathed / XLPE aluminium/copper conductor cable of 1.1 KV grade as required.							
a	4 Core 6 sq. mm XLPE ALAr	Nos	60	330	100	19,800	6,000	25,800.00
	TOTAL OF SUB HEAD II: LT CABLES					64,800	28,500	93,300.00
	Sub Head III - LIGHTNING ARRESTOR							
a	Supply & Installation of 8mm dia Aluminium alloy Solid Round Conductor of material used in Horizontal and down conductor system . Cross sectional area of conductor should be 50 mm ² .	Mtrs.	740	200	36	1,48,000	26,640	1,74,640.00
b	Supply & Installation of UV resistance Nylon/PU Conductor Holder for Parapet wall for holding 8 mm dia Aluminium alloy round Conductor	Nos	740	50	20	37,000	14,800	51,800.00
c	Supply & Installation of Nylon roof conductor holder for flat roof with concrete block for fixing 8 mm dia Aluminium alloy Solid Round Conductor in the terrace flat surface at every 1 mtr.	Nos	50	100	50	5,000	2,500	7,500.00
d	Supply & Installation of Stainless steel 304 Grade Cross Connector for 8 mm dia Aluminium alloy round conductor & 10 mm Copper bonded conductor at cross/Tee junction.	Nos	50	350	100	17,500	5,000	22,500.00
e	Supply & Installation of Expansion Joint with connector to compensate the expansion and contraction of Solid Round Al. Conductor during temperature variations. Expansion Joint should be consider at every 20 mtr length of straight horizontal air termination and connected at both the end with use of straight connector conductor.	Nos	38	300	50	11,400	1,900	13,300.00
f	Supply & Installation of Aluminium Straight conductor connector for interconnecting 8mm Aluminium conductor to meets the requirement of IS/IEC 62305.	Nos	26	50	20	1,300	520	1,820.00
g	Supply & Installation of air termination rod of 4mtr length of material AlMgSi of diameter ø 16/10 mm crimped on both the ends (Tapered Type). The rod shall be mounted with stainless steel clamp and fixing accessories . Test Parameter for 2 mtr vertical air terminal	Nos	11	8,000	1,150	88,000	12,650	1,00,650.00
	Down conductor							
h	Supply & Installation of 8mm dia Aluminium alloy Solid Round Conductor of material AlMgSi used in air termination and down conductor system . Cross sectional area of conductor should be 50 mm ² . Test Parameter: a. Lightning impulse current of 100kA for 10/350 µs b. Electrical resistivity - 0.034 µΩm, c. Tensile strength - 178 MPa or N/mm ² d. Salt spray - No Sign of Corrosion. Tested as per IEC 62561-2	Mtrs.	530	200	36	1,06,000	19,080	1,25,080.00

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS									Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT		
h	Supply & Installation of Nylon Conductor Holder for Parapet wall for holding 8 mm dia Aluminium alloy round Conductor.	Nos	530	50	20	26,500	10,600		37,100.00
i	Supply & Installation of Aluminium Straight conductor connector for interconnecting 8mm Aluminium conductor to meets the requirement of IS/IEC 62305. Test Parameters: a) Mechanical Load b) Electrical Resistivity c) Environmental Test Tested as per IEC 62561-1	Nos	36	50	20	1,800	720		2,520.00
j	Supply & Installation of Lightning counter - LCD screen shows the number of lightning strikes, hour and date of lightning events Buttons enable TIME/DATE setting and log viewing. Replaceable battery, working life minimum five years. Complies with IEC/EN 62561-6.	Nos	1	55,000	3,000				
i	Supply & Installation of Stainless Steel 304 grade Test Joint with enclosure to interconnect between 8 mm Al. round conductor to 10 mm copper bonded round conductor.	Nos	12	1,900	300	22,800	3,600		26,400.00
j	Supply & Installation of Maintenance Free Copper bonded Earth rod of 3048mm length having the dia of 25 mm Dia with copper coating thickness of 254 microns. The rod has been tested for Dimension, Marking, Tensile Strength, Salt mist, coating thickness, Electrical resistivity test before and after corrosion test as per	Sets	12	9,000	2,000	1,08,000	24,000		1,32,000.00
k	Supply & Installation of Rust resistant non corrosive 10 mm dia Copper Bonded Steel round conductor used for Earth termination inside the concrete and soil.	Mtrs.	65	500	100	32,500	6,500		39,000.00
	Total for SH III - LIGHTNING ARRESTOR					6,05,800	1,28,510		7,34,310.00
	Sub Head IV - Earthing								
1.0	Neutral Earthing with 17.2mm Dia 3Mtr. long Copper bonded steel rod including accessories and providing brick masonry work and enclosures with RCC slab and watering pipe etc., with Earth enhancing compound including excavation, refilling etc as required.	Each	6	15,000	2,500	90,000	15,000		1,05,000.00
2.0	Body Earthing with 40mm Dia 3 metre long, GI pipe including all accessories, and providing brick masonry work and enclosures with RCC slab and watering pipe etc. with Earth enhancing compound including excavation, refilling etc as required.	Each	20	8,000	2,500	1,60,000	50,000		2,10,000.00
3.0	Providing and fixing following sizes of GI / copper Flat / wire on surface or in recess for connections etc. as required.								
a	50 x 6 mm Copper Flat (Ti Neutral)	Meter	100	3,750	100	3,75,000	10,000		3,85,000.00
b	25 x 3mm Copper flat (250KVA DG Neutral)	Meter	50	2,700	50	1,35,000	2,500		1,37,500.00
c	50 x 6 mm GI Flat	Meter	400	250	75	1,00,000	30,000		1,30,000.00
d	25 x 6 mm GI Flat	Meter	200	145	50	29,000	10,000		39,000.00
e	25 x 3 mm GI Flat	Meter	600	100	30	60,000	18,000		78,000.00
f	8 SWG GI Wire	Meter	200	20	10	4,000	2,000		6,000.00
g	8 SWG Copper Wire	Meter	150	125	10	18,750	1,500		20,250.00
	Total for SH IV- Earthing					9,71,750	1,39,000		11,10,750.00
	Sub Head V- STREET /LANDSCAPE LIGHTING								
1.0	Supply, erection, testing and commissioning of Hot Dip Galvanised Octogonal Pole (with Door opening arrangement including supply of bakelite board with 5 A MCB, loop in and loop out connectors) of height 4 meters having bottom dimension of 130 mm and top dimension of 70 mm with base plate of 220 x 220 x 12 mm along with M20 700 mm long I Bolts with coil earthing using 80 turns of 8 SWG GI wire as per standards. The quoted rate shall also include the required foundation / RCC pedestal for mounting Street light pole with required foundation bolts.	Sets	20	8,000	2,000	1,60,000	40,000		2,00,000.00
2.0	Supply, erection, testing and commissioning of 0.5 meter long Single arm bracket made of 40/50 mm GI Pipe. On the above street light pole.	Each	20	1,500	500	30,000	10,000		40,000.00
3.0	Supplying and erecting of 45W LED street light fitting having PF > 0.95, class IP 65 or above Housing of pressure die cast aluminium alloy and heat sink extruded aluminium as per specification fixed on the Arm mentioned above for the street light pole.	Nos	20	3,000	1,200	60,000	24,000		84,000.00
4.0	Supply and Erection of Landscape lights with suitable drives for the following								
a	Post top lantern with 20W LED	Nos	5	5,500	750	27,500	3,750		31,250.00
b	Bollard fitting with 10W LED	Nos	5	4,500	750	22,500	3,750		26,250.00
c	Tree uplighter with 10W LED	Nos	5	1,500	500	7,500	2,500		10,000.00
5.0	Weather proof Junction Box for to accommodate 5Nos LED drives for above fitting with suitable provision for 4Cx6Somm PVC Al/Ar cable Terminations	Nos	3	8,000	750	24,000	2,250		26,250.00
	Total for SH V- STREET /LANDSCAPE LIGHTING					3,31,500	86,250		4,17,750.00
	Sub Head VI - SOLAR WORKS								
	Supply, installation, testing and commissioning of 15 kwvp ON-GRID Solar PV System / Plant consisting Solar PV Poly Crystalline Panel (540 Wp X 28 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / XLPO cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB, MCB etc. wiring up to solar Inverter in Tray/GI pipe, display board, sign board, two distinct earthing, lightning arrester, liaisoning charges, remote monitoring system with maintaining card/chip for 5 years including net charges for RMS System, complete up to net meter installation and export to the grid with 5 year ON-SITE warranty and fully comprehensive maintenance contract as per Specification no. ESD-RT-ONG	Set	1	11,25,000	75,000	11,25,000	75,000		12,00,000.00
	Total for SH VI - SOLAR WORKS					11,25,000	75,000		12,00,000.00
	Sub Head VII- MISCELLANEOUS WORKS								
1.0	Charges for preparation of drawings for HV & MV installations and obtaining approval of drawings, arranging inspection and obtaining safety certificate from Electrical Inspectorate excluding receiptable fee payable to them.	LS	1			1,50,000			1,50,000.00
2.0	Charges for Liasoning with supply authorities for obtaining load sanction and required power for the project excluding receiptable fee payable to them.	LS	1			2,50,000			2,50,000.00
3.0	Charges for preparation of 3sets AS BUILT drawings and maintenance manuals for major equipments for entire Electrical installation.	LS	1		50,000	-	50,000		50,000.00
4.0	Danger notice board Providing and installing danger notice board for medium voltage installation to be erected as per LS.	LS	4	750	150	3,000	600		3,600.00
5.0	Supply & Providing of printed instruction chart both in English and local language duly framed with front glasses for treatment of person suffering from Electric shock. of approved IS Make in the MV panel room	Sets	2	2,500	250	5,000	500		5,500.00
6.0	Supply & providing first aid chart duly framed and fixed in proper manner in MV panel room.	Sets	1	2,500	250	2,500	250		2,750.00
7.0	Supply & providing Co2 5Kg (Dry chemical) Fire extinguisher with necessary mounting bracket and fixed neatly in MV panel room/RMG Yard.	Each	4	8,500	350	34,000	1,400		35,400.00
8.0	Supply & providing fire buckets stand with 4 Nos. of fire buckets in the stand duly filled with sand. The stand shall be fabricated in the approved manner of the appropriate authorities and located outside MV panel room/RMG Yard.	Sets	2	5,000	250	10,000	500		10,500.00
9.0	Supplying and fixing PVC Synthetic Elastomer electrically Insulating Mat as per IS: 15652-2006 & CPRI tested of the following ratings.								
a	Class A insulation - 2 mm thick upto 3.3 KV	Sq Mtrs	20	1,500	100	30,000	2,000		32,000.00
b	Class B insulation - 2.5 mm thickness upto 11 KV	Sq Mtrs	5	2,500	100	12,500	500		13,000.00
c	Supply of 11 KV tested hand gloves.	Pair	2	2,500		5,000	-		5,000.00
	Total for SH VII : MISCELLANEOUS WORKS					5,02,000	55,750		5,57,750.00
	TOTAL FOR EXTERNAL ELECTRICAL WORKS					1,48,47,850	8,86,510		1,57,34,360.00
	III LIGHT FITTINGS AND FANS								
	Supply, installation, testing and Commissioning of following wattage LED Recessed / Surface mounting made of diecast aluminium body with powder coated/Extruded Plastic with Driver, operating voltage range of 150 to 265 Volts AC, Power factor > 0.95 high power 1 watt having efficacy of > 100 lumens/watt, 6500K, >80 CRI, 120deg beam angle, IP20, 3KV, <15% THD etc., complete with 3 years warranty.								
1									
a	1X18Watt LED Bulk head fitting - Pressure diecast housing with robotic polycarbonate cover, IP65, IK09 etc.,	Nos	38	650	150	24,700	5,700		30,400.00
b	1X20Watt LED Batten type fitting - Extruded Plastic with white acrylic cover.	Nos	15	150	150	2,250	2,250		4,500.00

PROPOSED CONSTRUCTION OF 122 FLATS AT DN NAGAR, ANDERI WEST, MUMBAI- PART F: BOQ FOR ELECTRICAL WORKS									Total Amount including supply and installation
SL NO	DESCRIPTION	UNIT	QTY	SUPPLY RATE	ERECTION RATE	SUPPLY TOTAL AMOUNT	ERECTION TOTAL AMOUNT		
c	1X10 Watt LED Mirror fitting - Extruded Plastic with white acrylic cover.	Nos	306	165	150	50,490	45,900		96,390.00
d	Signage light fitting with 30min battery backup.	Nos	90	5,500	200	4,95,000	18,000		5,13,000.00
e	Wall Light fitting with 18W LED Extruded Plastic with white acrylic cover	Nos	157	175	150	27,475	23,550		51,025.00
f	Ceiling light with 18W LED with Motion sensor Extruded Plastic with white acrylic cover(For Toilet)	Nos	306	900	200	2,75,400	61,200		3,36,600.00
g	Ceiling surface square downlighter with 18W LED Extruded Plastic with white acrylic cover	Nos	1813	650	150	11,78,450	2,71,950		14,50,400.00
h	Ceiling surface circle downlighter with 18W LED Extruded Plastic with white acrylic cover	Nos	1612	600	150	9,67,200	2,41,800		12,09,000.00
i	Chandelier Light fitting with 5 LED lamps	Nos	126	4,500	350	5,67,000	44,100		6,11,100.00
j	Night Lamp with 3W LED (Flush type)	Nos	306	600	200	1,83,600	61,200		2,44,800.00
k	Recess light fitting with 18W LED Extruded Plastic with white acrylic cover.	Nos	12	600	200	7,200	2,400		9,600.00
l	Wall light fitting with 18W LED (Weather proof) Extruded Plastic with white acrylic cover.	Nos	20	2,000	150	40,000	3,000		43,000.00
m	50W LED Flood light weather proof	Nos	10	1,900	350	19,000	3,500		22,500.00
n	2'x2' Surface fitting with 36W LED	Nos	220	2,000	200	4,40,000	44,000		4,84,000.00
2.0	AVIATION LIGHT FIXTURE								
a	Supply and fixing of aviation obstruction light of 1 No. fitted in single arm with LED lights 12 Watts with built in power supply on 1 meter long 50 mm dia GI Pipe with all accessories as required.	Nos	2	4,800	750	9,600	1,500		11,100.00
3	CALLING BELL								
a	Supplying and erecting Electronic type cordless call bell erected on T.W. board having portable bell push working on 9V D.C. battery and (effective range upto 45 meter) bell on 250 V.A.C. supply.	Nos	126	900	150	1,13,400	18,900		1,32,300.00
4	CEILING / EXHAUST FAN								
a	Ceiling Fan - BLDC Type (1200mm)	Nos	684	3,000	300	20,52,000	2,05,200		22,57,200.00
b	Exhaust fan (300mm)	Nos	432	1,400	250	6,04,800	1,08,000		7,12,800.00
5	GEYSER								
a	Geyser - 25 Litres (3 star rated)	Nos	306	7,500	750	22,95,000	2,29,500		25,24,500.00
	TOTAL FOR LIGHT FITTINGS AND FANS					93,52,565	13,91,650		1,07,44,215.00
IV	DG SET & ALLIED WORKS:								
1	DG SET								
	Supply,Erection, testing and commissioning of factory assembled Prime Duty Diesel Generating set as per CPCB IV norms comprising of 1500 RPM, 415 Volts , 50 Hz 0.8 Power Factor Alternator Coupled to Radiator cooled Diesel Engine of following capacity with Micro Processor based DG Control panel complete with								
	Base Frame Made of MS Channel including applicable painting.								
	Vibration isolation with Spring Isolators .								
	990ltr Inbuilt diesel tank as per standards with Level Indicators,Fuel piping and other necessary accessories as per standards.								
	Dry Exhaust manifold with Residential Grade silencer and stainless steel flexible connectors								
	Required cladding for heat insulation should be considered as part of quoted rate.								
	24 Volt starting system comprising of Starter Motor, Voltage Regulator, charged Lead Acid batteries of suitable ampere hour as per manufacturer's standard								
	Acoustic and Weather proof enclosure with arrangement for fresh air intake for cooling of Engine & Alternator, extraction and discharge hot air to atmosphere as per detailed specification.								
	Micro Processor Based ENGINE CONTROLLER SUITABLE FOR Auto start / stop								
	The DG Set shall confirm to attached Particular & Detailed specification.								
	The electrical control panel located along with the DG set and to be provided with the acoustic enclosure shall consist of suitable ACB								
	The required exhaust pipe upto the terrace floor with relevant cladding and MS support arrangement for fixing the Exhaust pipe from the DG set up to terrace as mentioned below in the Item 2.0.								
	The quoted rate here should consists of relevant Bird mesh at the top end of the Exhaust pipe at terrace or wherever it ends								
1.1	250KVA/ 200 KW	Each	1	25,00,000	55000	25,00,000	55000		25,55,000.00
2	EXHAUST PIPE								
2.1	Supply , installation , testing and commissioning of following size MS Exhaust piping of class ' B ' of following sizes thermally insulated with 50 mm thick mineral wool insulation with chicken mesh lagging and 24 SWG Aluminium Cladding .								
	The DG exhaust pipe should be taken up to 3mtrs from terrace floor situated above 8 th floor (as per CPCB norms).								
	The exhaust pipe should painted with 2 coats of suitable heat resistant Aluminium paint after preliminary and good cleaning.								
a.	6 inch dia	Meter	45	9,500	950	4,27,500	42750		4,70,250.00
2.2	Supply and Installation of suitable MS supports for above exhaust pipe with necessary hardwares for horizontal run in ground and vertical in wall as per drgs.	Kgs	5000	120	25	6,00,000	125000		7,25,000.00
	TOTAL FOR DG SET AND ALLIED WORKS					3527500	222750		37,50,250.00
V	CCTV AND ALLIED WORKS								
A	S.I.T.C of IP CCTV system as under :Uniview/Hikvision/Dahua)								
1	Supply, Installation, Testing and Commissioning of UL Listed 1/3" 2MP CMOS Dome type camera with 2.8 mm Fixed lens, Minimum illumination: Color : 0.1Lux and BW: 0Lux (IR LED on), 25 fps at 1920 x 1080 , Progressive scan, 120dB WDR or better, True Day & Night, IR minimum distance of 20 Mtr, Motion detection, Audio In/ Out, PoE class3 and Power Supply 12V DC ±10% Port.	Nos	40	6,000	600	2,40,000	24,000		2,64,000.00
2	Supply, Installation, Testing and Commissioning of UL Listed 1/3" 4MP CMOS Bullet type outdoor camera (IP65) with 2.8 - 12 mm Varifocal lens, Minimum illumination: Color : 0.1Lux and BW: 0Lux (IR LED on), 25 fps at 1920 x 1080 , Progressive scan, 120dB WDR or better, True Day & Night, IR minimum distance of 20 Mtr, Motion detection, Audio In/ Out, PoE class3 and Power Supply 12V DC ±10% Port.	Nos	14	8,000	900	1,12,000	12,600		1,24,600.00
	NETWORKING COMPONENTS								
3	Supply, Installation, Testing and Commissioning of CAT6 I/O point with Face plate and MS mounting box(powder coated) and necessary fixing accessories, (complete set)	Nos	108	1,750	600	1,89,000	64,800		2,53,800.00
4	Supply, Installation, Testing and Commissioning of 8 port Gigabit Layer2 PoE/POE + switch (370W Budget Power) with 4x1G SFP+ uplink ports with 1000BASE-LX SFP optic, SMF Module, LC connector, and direct attached 1G SFP Copper Cable 1 M Stacking cable as per the specification. (AXM762F 10Gig SFP+ module) (Make: D-Link)	Nos	3	1,69,000	7,500	5,07,000	22,500		5,29,500.00
5	Supply, Installation, Testing and Commissioning of 6 port Gigabit Layer2 PoE/POE + switch (370W Budget Power) with 4x1G SFP+ uplink ports with 1000BASE-LX SFP optic, SMF Module, LC connector, and direct attached 1G SFP Copper Cable 1 M Stacking cable as per the specification. (AXM762F 10Gig SFP+ module) (Make: D-Link)	Nos	7	1,69,000	7,500	11,83,000	52,500		12,35,500.00
6	Supply, Installation, Testing and Commissioning of 4 port Gigabit Layer2 PoE/POE + switch (370W Budget Power) with 4x1G SFP+ uplink ports with 1000BASE-LX SFP optic, SMF Module, LC connector, and direct attached 1G SFP Copper Cable 1 M Stacking cable as per the specification. (AXM762F 10Gig SFP+ module) (Make: D-Link)	Nos	1	1,69,000	7,500	1,69,000	7,500		1,76,500.00
7	Supply, Installation, Testing and Commissioning of 24 port layer 2 switch with 10/100/1000 POE Managed Switches with IP multicast snooping and data-driven IGMP support and with two 1000baseT single mode FO uplink ports (with ready Fiber Optic module on Board) and all related termination accessories(Like Pigtails/UUs/Convertors/Splicing equipment Etc.) as per site specific System requirement. (AXM762F 10Gig SFP+ module) (Make: D-Link)	Nos	3	2,29,000	9,000	6,87,000	27,000		7,14,000.00
	NVR								
8	Supply, Installation, Testing and Commissioning of 64 channel NVR - networkable video real time recorder , D1 Resolution DVR , H.264 compression, 64ch video input,1ch audio input (ICA),8 X 8 SATA HDD support, USB 2.0,Record: support D1 real-time recording, dual encoding streams, smart phone monitoring etc with HDMI port , Min 128Mbps/128Mbps transfer and recording speed,RJ-45 1000 Base-T ethernet port, DDNS support, and with required power supply accessories. (Industrial type (24*7) SATA hard disk for 60 days, 4CIF recording for 60 camera)	Nos	1	1,08,000	8,000	1,08,000	8,000		1,16,000.00
9	Supply, Installation, Testing and Commissioning of Workstation as per the specification:- HP Victus 15L Desktop AMD Ryzen 5 5600G (8GB RAM Upto 32GB RAM/512GB SSD/AMD Radeon RX 6400 Graphic /Win11/MSO/XBOX Game Pass/Mica Silver) TG02-0102in 1GB Ethernet Card, Microsoft Windows 7 (64 bit), 1 No. 32" LED Monitor, Keyboard, Mouse with1.No. of 10m length HDMI cable and accessories	Set	1	5,58,000	7,500	5,58,000	7,500		5,65,500.00

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